Master Thesis

Demand Estimation On Amazon Products Using The BLP Approach

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Abstract

This thesis employes the Berry, Levinsohn, and Pakes (BLP) model to estimate demand and evaluate market dynamics within the UK and DE beauty product sectors on Amazon. Utilizing the pyBLP package, the study analyzes price sensitivity, cross-price elasticities, and diversion ratios, providing insight into consumer preferences and competitive structures. A hypothetical merger scenario between NIVEA and CeraVe is also investigated, revealing significant post-merger price increases and market power implications. The findings suggest consumer price sensitivity and identify potential substitutes and complements within the market.

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1 Introduction

Demand elasticities and market shares are crucial concepts in understanding consumer behavior and market dynamics. The estimation of demand plays a central role in various economic analyses, including the assessment of market power, mergers, innovation, and the valuation of new brands in differentiated-products markets [Nevo, 2000]. This estimation is essential for understanding how consumers respond to changes in prices, incomes, and other factors, and how these responses impact market shares and the overall competitiveness of industries.

Recent studies have highlighted the significance of demand estimation in modern economies. For instance, research on mobile applications has demonstrated the substantial impact of estimated demand functions on consumer surplus and market outcomes [Ghose and Han, 2014]. Moreover, empirical models of demand and supply in differentiated products industries have provided valuable insights into the interplay between consumer preferences, product differentiation, and overall welfare [Gandhi and Nevo, 2021]. These kinds of analyses highlight the significance of understanding the demand dynamics across many different sectors from technology to food and beverages.

Furthermore, the use of advanced econometric techniques, such as random-coefficient logit models and generalized method of moments (GMM) estimation, has enabled researchers to capture complex consumer choices and market behaviors from the analysis of new product introductions in the yogurt market to the examination of demand for various goods and services as said by [Giacomo, 2007] and [Poonyth et al., 2001]. These methodologies offer a clear view of demand estimation, relevant across different products and industries.

Various methodologies and empirical models have been developed to estimate demand elasticities and market shares to provide insights into consumer behavior and market dynamics. One approach involves the use of random-coefficients logit models, which have been employed to estimate demand for differentiated products, taking into account zero-valued market shares [Dubé et al., 2020].

Furthermore, empirical models of demand and supply in differentiated products industries have been developed to understand the complex interplay between consumer preferences and market dynamics [Gandhi and Nevo, 2021]. These models offer a comprehensive framework for estimating demand elasticities and market shares, shedding light on the intricate relationship between product differentiation and consumer behavior.

The demand elasticities studies conducted in specific markets by [Yen, 2003], [Calatrava, 1990], and [McCamley and Kliebenstein, 1981], such as the European avocado market, cigarettes, and diesel fuel, showcase the broad spectrum of applications for these methods. These diverse contexts highlight the universal relevance of demand elasticity analysis, ranging from public goods to specific consumer prod-

ucts [Yen, 2003], [Reimers, 1976].
Demand elasticities and market shares on Amazon products are crucial topics in today's e-commerce landscape. As the markets engineered by Amazon are constantly evolving [Rogers, 2017], understanding demand elasticities is essential for businesses to make informed decisions. Additionally, insights from studies on organic milk demand and supply highlight the significance in understanding consumer behavior [Su,].
The e-commerce landscape, particularly within the realm of Amazon products, is a dynamic and complex environment that necessitates a comprehensive understanding of demand elasticities and market shares. The demand for differentiated products in such industries is a critical area of study [Gandhi and Nevo, 2021], as it directly impacts market dynamics and consumer behavior. The estimation of demand, especially in the presence of zero-valued market shares, is a significant challenge that requires advanced statistical techniques [Dubé et al., 2020].
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The utilization of the random coefficient logit model in analyzing demand elasticities and market shares on Amazon products presents a sophisticated and robust approach to econometric modeling. This model allows for the incorporation of varying individual preferences and characteristics, providing a more nuanced understanding of consumer behavior [Swamy and Tavlas, 1995]. The random coefficient logit model has gained prominence due to its ability to address endogeneity and accommodate datasets with zero-valued market shares, making it particularly suitable for analyzing consumer choices in the e-commerce domain [Dubé et al., 2020]. Furthermore, the application of random coefficient models has been instrumental in understanding changes in consumption preferences, as evidenced by studies in diverse contexts such as South Africa's meat consumption patterns [Poonyth et al., 2001].
This thesis provide a comprehensive analysis and delves into a focused and detailed investigation: What are the demand elasticities and market shares of skincare products on Amazon UK and Amazon DE, and how are they influenced by product characteristics such as ratings, current price, previous month sales, reviews, and specific product features? Utilizing a unique dataset comprising 88 common skincare creams from these two Amazon platforms, this research aims to meticulously estimate and analyze the nuances of demand elasticity within this

specific segment of the e-commerce market. The study will explore how these various product characteristics impact consumer purchasing decisions and overall market dynamics on these platforms. This analysis not only seeks to provide a deep understanding of consumer behavior in the skincare product category but also aims to offer insights into the broader patterns of e-commerce market trends in the UK and German markets. By focusing on these specific aspects and regions, the research endeavors to contribute to the field of economic analysis in e-commerce, offering valuable implications for businesses, marketers, and policymakers engaged in the online retail of skincare products.

Following this introduction, the paper is structured as follows: Section 2, the Literature Review, delves into the foundational concepts underpinning our study, including market shares, utility theory, demand elasticities, and discrete choice models and with an examination of the Amazon marketplace. Section 3, Theoritical Framework, outlines the analytical models used in our research, plain logit, random coefficient logit and nested logit models, and discusses their application in economic analysis. Section 4 details the Data Collection and Preprocessing, highlighting variable selection and data cleaning processes. Section 5 presents the sentiment analysis performed on the product's reviews. We continue with section 6 where is presented the methodology we used to gain our results. We outline the whole procedure from the data preprossesing to applyt he BLP model, to the estimation of market shares, the creation of the BLP-style instruments, and finally the application of our model. In section 7 we present the empirical results and also the post-estimation analysis on our models. We conclude with section 8 with the conclusions of our research.

2 Literature Review

2.1 Market Shares

In this section of my research, I will introduce you to the significant concept of the market shares and their significance in diverse industries. Market share estimation is a fundamental aspect of demand analysis and provides valuable insights into consumer behavior, market dynamics, and competitive strategies. It serves as a cornerstone for developing robust demand estimation models and enables businesses to make data-driven decisions to meet consumer needs and achieve competitive advantage. The concept of market shares is essential for understanding the competitive landscape and strategic positioning of firms [Ghosh, 2004]. As market share increases, the quantity of products being sold also increases, enabling firms to attain economies of scale that lead to low-cost production [Kim et al., 2020].

2.1.1 Definitions and Industry Applications

Estimating market shares is crucial for understanding and predicting demand in various industries. Market share estimation provides valuable insights into consumer preferences, brand performance, and competitive dynamics, which are essential for making informed business decisions [Nevo, 2000]. By analyzing market shares, firms can assess their relative position in the market and evaluate the effectiveness of their marketing strategies. Additionally, market share estimation is instrumental in assessing market power, mergers, and the introduction of new products, as it provides a quantitative understanding of consumer behavior and preferences [Gayle and Lin, 2022].

Furthermore, market share estimation plays a vital role in demand modeling. It allows for the identification of consumer preferences and willingness-to-pay for different products, which are essential components of demand estimation models. Understanding market shares enables businesses to make informed pricing decisions, optimize product portfolios, and design effective marketing strategies to meet consumer demand. Moreover, market share estimation is integral to assessing the impact of new product introductions on consumer preferences and market dynamics [Dubé et al., 2012].

Market shares represent the portion of total sales or revenue that a company or product captures within a specific market. It is a crucial metric for assessing a firm's competitive position and performance in relation to its industry peers [Kim et al., 2020]. Market shares are widely used in various industries as a key performance measure, particularly in consumer packaged goods markets [Tanusondjaja et al., 2020].

2.1.2 Strategic And Economic Importance Of Market Shares

The importance of market shares lies in their ability to provide insights into a firm's competitive advantage, profitability, and growth potential
[Kim et al., 2020]. Additionally, market shares play a significant role in influencing stock prices, as evidenced by the internal financial determinants of stock prices in the banking sector [Azmeh and Hamada, 2022]. Moreover, increasing market share is often considered a rationale for corporate acquisitions, highlighting its strategic significance in business decisions [Ghosh, 2004]. Importantly, market shares are used as a basis for assessing the impact of exchange rate pass-through on firms' market shares, demonstrating their relevance in international trade and economics [Devereux et al., 2015].
In the context of differentiated products, market share estimation is particularly valuable for understanding the price effects of mergers, the value of new goods, and the contribution of products to seller networks. Differentiated product demand systems are widely used for analyzing the competitive landscape and evaluating the implications of various market scenarios on consumer choices and market outcomes [Conlon and Gortmaker, 2020].
The applications of market shares extend beyond traditional business contexts. For instance, market share mobility in the newspaper industry embodies the level of concentration and its change, as well as the correlation of market shares over time, reflecting its relevance in media economics [Kranenburg, 2002]. Furthermore, market shares are utilized in the context of collective reflexivity in social marketing, aiming to extend the concept of collective reflexivity into marketing theory [Kariippanon et al., 2019]. The concept of market shares is also employed in the context of the sharing economy, providing a marketing perspective on this emerging economic model [Lim, 2020].
In summary, market shares are a fundamental metric for evaluating firms' competitive positions, strategic decision-making, stock prices, and industry dynamics. Their applications span across various industries and disciplines, making them a critical aspect of business analysis and strategic planning.

2.2 Theory Of Utility And Demand

To introduce the theory of utility and demand, it is essential to understand consumer behavior and the estimation of demand. Utility theory plays a fundamental role in this understanding, as it provides insights into consumer preferences and decision-making processes. Several studies have focused on estimating demand and utility, offering valuable insights into consumer behavior and market dynamics.

2.2.1 Fundamentals Of Utility Theory And Demand

The fundamentals of utility theory and demand are crucial for understanding consumer behavior and market dynamics. Utility theory provides insights into consumer preferences and decision-making processes, which form the basis for subsequent discussions on demand and elasticities [Berry, 1994].
One proposed approach involves estimating mean levels of utility for each product by 'inverting' the market share equation, providing a method for utility level estimation [Nevo, 1998]. This method underscores the importance of estimating demand parameters, particularly in assessing market power, as exemplified in the ready-to-eat cereal industry. It emphasizes the role of demand estimation in unraveling market dynamics.
Moreover, research has highlighted the significance of estimating taste parameters for product characteristics, revealing variations in consumer preferences across different products [Knittel and Metaxoglou, 2014]. Additionally, documented challenges exist in estimating random-coefficient demand models, shedding light on the complexities inherent in demand estimation [Reynaert and Verboven, 2014].
To address these complexities and enhance the performance of random-coefficient demand models, discussions center around the role of optimal instruments, underlining the necessity for methodological advancements in demand estimation.
In exploring the concept of demand estimation and utilities, it is important to advise the perspective offered by the documentation of
PyBLP [PyBLP Documentation, Year].
Estimating demand relies on understanding consumer preferences and their decision-making processes. Utilities play a significant role in this estimation process. Utilities represent the satisfaction or preference that consumers derive from consuming different goods and services. In the context of demand estimation, utilities are essential for modeling consumer choice behavior. The random-coefficients logit models of demand, as discussed by [PyBLP Documentation, Year], emphasize the importance of understanding the heterogeneity in consumer preferences, which is captured through the utility function. These models recognize that consumers have varying preferences and willingness-to-pay for products, and utilities serve as a fundamental component in capturing this heterogeneity.
Observed demand-side product characteristics are contained in the $N \times K_1$ matrix of linear characteristics, X_1, and the $N \times K_2$ matrix of nonlinear characteristics, X_2, which is typically a subset of X_1. Unobserved demand-side product characteristics, ζ, are a $N \times 1$ vector.

In market t, observed agent characteristics are an $I_t \times D$ matrix called demographics, d. Unobserved agent characteristics are an $I_t \times K_2$ matrix, ν.

The indirect utility of agent i from purchasing product j in market t is given

by:

$$U_{ijt} = \underbrace{\delta_{jt} + \mu_{ijt}}_{V_{ijt}} + \varepsilon_{ijt}$$

In which the mean utility is, in vector-matrix form:

$$\delta = \underbrace{X_1^{en}\alpha + X_1^{en}\beta^{ex} + \xi}_{X_1\beta}.$$

The $K_1 \times 1$ vector of demand-side linear parameters, β, is partitioned into two components: α is a $K_1^{en} \times 1$ vector of parameters on the $N \times K_1^{en}$ submatrix of endogenous characteristics, $X_1^{en}, and \beta^{ex}$ is a $K_1^{ex} \times 1$ vector of parameters on the $N \times K_1^{ex}$ submatrix of exogenous characteristics, X_1^{ex}. Usually, $X_1^{ex} = p$, prices, so α is simply a scalar.

The agent-specific portion of utility in a single market is, in vector-matrix form:

$$\mu = X_2 \left(\Sigma\nu' + \Pi d'\right).$$

The model incorporates both observable (demographics) and unobservable taste heterogeneity through random coefficients. For the unobserved heterogeneity, we let ν denote independent draws from the standard normal distribution. These are scaled by a $\mathbf{K}_2 \times \mathbf{K}_2$ lower-triangular matrix Σ, which denotes the Cholesky root of the covariance matrix for unobserved taste heterogeneity. The $\mathbf{K}_2 \times \mathbf{D}$ matrix Π measures how agent tastes vary with demographics.

In the above expression, random coefficients are assumed to be normally distributed, but this expression supports all elliptical distributions. To incorporate one or more lognormal random coefficients, the associated columns in the parenthesized expression can be exponentiated before being pre-multiplied by $\mathbf{X}_2$. For example, this allows for the coefficient on price to be lognormal so that demand slopes down for all agents. For lognormal random coefficients, a constant column is typically included in $\mathbf{d}$ so that its coefficients in Π parametrize the means of the logs of the random coefficients. More generally, all log-elliptical distributions are supported. A logit link function is also supported.

Random idiosyncratic preferences, ε_{ijt}, are assumed to be Type I Extreme Value, so that conditional on the heterogeneous coefficients, market shares follow the well-known logit form. Aggregate market shares are obtained by integrating over the distribution of individual heterogeneity. They are approximated with Monte Carlo integration or quadrature rules defined by the $\mathbf{I}_t \times \mathbf{K}_2$ matrix of integration nodes, ν, and an $\mathbf{I}_t \times 1$ vector of integration weights, $\mathbf{w}$:

$$s_{jt} \approx \sum_{i \in \mathbf{I}_t} w_{it} s_{ijt},$$

where the probability that agent i chooses product j in market t is

$$s_{ijt} = \frac{\exp(V_{ijt})}{1 + \sum_{k \in \mathbf{J}_t} \exp(V_{ikt})}.$$

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where the probability that agent i chooses product j in market t is

$$s_{ijt} = \frac{\exp(V_{ijt})}{1 + \sum_{k \in \mathbf{J}_t} \exp(V_{ikt})}.$$

The one in the denominator arises because the utility of the outside good is normalized to $U_{i0t} = 0$. The scale of utility is normalized by the variance of ε_{ijt}. Moreover, [Gayle and Lin, 2022] highlight the impact of new product introductions on consumer preferences and willingness-to-pay. The estimation of demand for new products relies on understanding how the utility derived from these products influences consumer choices. As consumers' relative preference and willingness-to-pay for new products evolve over time, utilities play a crucial role in capturing these changes and incorporating them into demand estimation models.

2.2.2 Applying Utility Concepts to Consumer Behavior

In essence, utilities are pivotal in demand estimation as they encapsulate the underlying preferences and satisfaction that drive consumer choices. By incorporating utilities into demand models, researchers and businesses can gain insights into consumer behavior, assess the impact of new products, and make informed decisions regarding pricing, product development, and market strategies.
Applying utility concepts to consumer behavior involves understanding the intricacies of demand estimation and its implications for market dynamics. [Berry, 1994] proposed estimation by "inverting" the market share equation to find the implied mean levels of utility for each good, providing a method for estimating utility levels. This approach is discussed in more detail by [Nevo, 1998], who highlighted the exploitation of time variation in weekly prices to identify demand parameters. Nevo's work emphasizes the practical application of utility concepts in measuring

market power. These insights underscore the significance of demand estimation in understanding market dynamics. Additionally, [Chidmi and López, 2007] emphasized the significance of estimating taste parameters for product characteristics, indicating the variability of consumer preferences across different products. These references collectively underscore the practical application of utility concepts in understanding consumer behavior and market dynamics.
Furthermore, [Reynaert and Verboven, 2014] discussed the role of optimal instruments in improving the performance of random coefficients demand models, addressing the need for methodological advancements in demand estimation. They provided insights into improving the numerical performance of static and dynamic aggregate discrete choice random coefficients demand estimation, highlighting the ongoing efforts to enhance the accuracy and efficiency of demand estimation techniques [Dubé et al., 2012].
A study by [Xiong et al., 2023] demonstrated the application of the random coefficient logit model in exploring consumer preferences for electric vehicles. In this context, utilities are essential in understanding the factors that drive consumer choice towards electric vehicles, such as subsidies, battery technology, and other attributes. The model allows for the estimation of utilities associated with these attributes, providing insights into their impact on consumer demand and market shares.
In summary, utilities are essential in estimating demand using models such as the random coefficient logit model. They serve as the foundation for understanding consumer preferences, making accurate predictions of demand, and ultimately estimating market shares. By incorporating random coefficients and addressing endogeneity, this model provides a robust framework for capturing utilities and effectively estimating demand in markets with differentiated products.

2.3 Demand Elasticities: Definition and Importance

Demand elasticity refers to the responsiveness of the quantity demanded of a good or service to changes in its price, income, or the price of related goods. It is a crucial concept in economics as it helps in understanding consumer behavior and market dynamics. Price elasticity of demand, income elasticity, and cross-price elasticity are the main types of demand elasticity [Rusli and Kamu, 2020]. Price elasticity of demand measures the percentage change in quantity demanded in response to a one percent change in price. Income elasticity measures the percentage change in quantity demanded in response to a one percent change in income, while cross-price elasticity measures the percentage change in quantity demanded of one good in response to a one percent change in the price of another good [Kyer and Maggs, 2015].
Several factors affect demand elasticity. For instance, the availability of substi-

tutes influences the price elasticity of demand, with goods having close substitutes being more elastic [Vukadinović et al., 2017]. Additionally, the time horizon is a crucial factor, as demand tends to be more elastic in the long run compared to the short run [Vives and Jacob, 2023]. Moreover, the nature of the good or service, such as whether it is a necessity or a luxury, affects its demand elasticity. Necessities tend to have inelastic demand, while luxuries have elastic demand [Bielik and Šajbidorová, 2009]. Furthermore, consumer preferences, market competition, and the level of advertising also impact demand elasticity [Kinnucan and Zheng, 2004].
In conclusion, demand elasticity is a fundamental concept in economics, providing insights into consumer behavior and market dynamics. Price elasticity of demand, income elasticity, and cross-price elasticity are essential measures in understanding how demand responds to changes in price, income, and the prices of related goods. Various factors, including the availability of substitutes, time horizon, nature of the good, consumer preferences, market competition, and advertising, play significant roles in influencing demand elasticity.

2.3.1 Estimation of Demand Elasticities

Estimating demand elasticities is a fundamental aspect of empirical demand analysis. Various methods have been developed to estimate demand elasticities, particularly for differentiated products. One common approach is the use of random-coefficients logit models, which allow for flexibility in capturing unobserved consumer heterogeneity and product differentiation [Nevo, 2000]. These models are estimated using econometric techniques such as the Berry–Levinsohn–Pakes (BLP) estimator, which has been widely used in the literature to estimate demand for differentiated products [Vincent, 2015]. Additionally, the use of instrumental variables and generalized method of moments (GMM) estimation has been advocated to address endogeneity and improve the performance of demand models [Wooldridge, 2001]. Furthermore, the estimation of demand elasticities often involves addressing challenges such as the large number of elasticities to estimate and the correlation between prices and unobserved product characteristics, which can bias the elasticity estimates [Berry, 1994][Berry et al., 1995].
Moreover, recent studies have emphasized the importance of considering discrete sources of market segmentation and accounting for continuous product characteristics when estimating demand elasticities
[Grigolon and Verboven, 2014]. The estimation process also involves exploiting time-variation in prices to identify demand parameters and obtaining initial parameter estimates for implementation as said by [Nevo, 1998] and
[Reynaert and Verboven, 2014]. It is crucial to note that the estimation of demand elasticities provides valuable insights into consumer behavior, market power, and

the effects of mergers and new product introductions. Overall, the estimation of demand elasticities is a complex yet essential task that requires careful consideration of econometric techniques, addressing endogeneity, and accounting for product differentiation and consumer heterogeneity.

2.3.2 Role of Elasticities in Market Analysis

Market elasticities play a crucial role in market analysis by providing insights into consumer behavior, pricing strategies, and market equilibrium. Elasticities, such as own-price elasticities and cross-price elasticities, are essential in understanding how changes in prices affect the quantity demanded and the relationships between different products in the market [Berry et al., 1995]. These elasticities are particularly important in estimating demand models and assessing market power, as they provide valuable information on consumer preferences and the responsiveness of demand to price changes [Nevo, 2000]. Furthermore, market elasticities are vital for evaluating the impact of mergers on consumer welfare and market competition knittel2014.
In practical applications, market elasticities have been utilized to examine consumer choices, retail competition, and brand-level demand for various products, including breakfast cereals and coffee
[Chidmi and López, 2007][Szymanski, 1993]. Furthermore, the relationship between market share and profitability has been a subject of interest, with implications for strategic decision-making by firms aiming to increase their market share.

2.4 Amazon Marketplace

The growth of Amazon Marketplace and other digital marketplaces has been substantial, shaping the retail industry, influencing global trade, and presenting opportunities and challenges for various stakeholders within the marketplace ecosystem. The expansion of multi-sided marketplaces, including Amazon, Alibaba, eBay, JD.com, and Rakuten, has been a key driver of growth in the global retail sector, as evidenced by their substantial annual revenue and gross merchandise value (GMV) [Hänninen et al., 2019]. Retailers on the marketplace, such as Amazon, have been indicative of growth opportunities for the marketplace operators, underscoring the mutually beneficial relationship between the platform and its sellers [Maier and Wieringa, 2021].
The impact of online referral services and the participation of branders in marketplace activities has also contributed significantly to the growth of both branders and platforms, highlighting the collaborative nature of growth within the marketplace ecosystem [Chu et al., 2023].
Additionally, the dynamics of growth within online platforms, such as Amazon

Marketplace, have been influenced by contractual design considerations, particularly regarding pricing strategies and commission fees, reflecting the evolving nature of marketplace operations and their impact on vendor relationships [Chen et al., 2022]. The paradox of market reach and financial performance on digital platforms, as well as the pressures faced by independent sellers in accessing global demand while navigating competitive challenges, further underscores the complexities and nuances of growth within the digital marketplace landscape [Hänninen and Smedlund, 2021].

The significance of the Amazon Marketplace in the cosmetic and beauty sector is notable due to its impact on consumer behavior, market dynamics, and industry trends. The marketplace's dominance and growth have influenced the cosmetic and beauty industry in various ways. Amazon's role as a dominant partner in the marketplace system has led to increased competitiveness against other competitors, affecting segments within the global cosmetic and beauty markets [Ryan et al., 2012].

Price dispersion observed on Amazon Marketplace, particularly in the prices charged by third-party sellers, reflects the marketplace's influence on pricing strategies within the industry [Mantin et al., 2014]. The impact of the transformation and reconfiguration of the beauty industry, particularly in the context of cosmetic tourism, has also been associated with the development of soft power and nation branding strategies, highlighting the broader influence of market trends on national and global perceptions of beauty industries [Lim, 2020].

3 Theoritical Framework

3.1 The BLP Model

The BLP (Berry, Levinsohn, and Pakes) model is a widely used framework in industrial organization and empirical IO. It is a structural model of demand and supply that allows for product differentiation and firm heterogeneity. The BLP model has been extensively applied in various studies to estimate demand and supply parameters in differentiated-products markets [Nevo, 2000]. This model has been particularly valuable in examining questions of market power, mergers, innovation, and valuation of new brands [Nevo, 2000]. The BLP model's flexibility and ability to account for unobserved product characteristics and consumer preferences make it a powerful tool for understanding market dynamics and consumer behavior [Conlon and Gortmaker, 2020].

The BLP can provide some tools to estimate different types of models, such as Logit, Nested Logit and Random Coefficients Nested Logit models. There is a library in python called PyBLP, the one is used in this research, that also includes some features for dealing with the fixed effects and instrumental variables. This library is designed for applications in empirical industrial organization (IO) and focuses on market stracture and competition analysis [PyBLP Documentation, Year].

The BLP model has also been used to estimate demand for specific products, such as electric vehicles . Its application has provided valuable insights into consumer preferences and the factors influencing the demand for electric vehicles, which is essential for policy-making and subsidy allocation in the energy sector [Xiong et al., 2023]. Furthermore, the BLP model has been employed to estimate demand for mobile applications, revealing significant enhancements in consumer surplus and offering implications for mobile marketing analytics, app pricing, and design strategies [Ghose and Han, 2014].

Overall, the BLP model represents a significant advancement in the estimation of demand and supply parameters in differentiated-products markets, providing a robust framework for understanding consumer behavior, market competition, and the effects of new product introductions.

The BLP (Berry, Levinsohn, and Pakes) model encompasses the random-coefficients logit model and its extensions, which have been instrumental in estimating demand for differentiated products. The random-coefficients logit model is a fundamental component of the BLP framework, allowing for the estimation of demand while considering product differentiation and consumer heterogeneity [Nevo, 2000].

Furthermore, the extensions of the random-coefficients logit model within the BLP framework have facilitated a more comprehensive analysis of consumer preferences and market competition. These extensions have allowed for the estimation of models that account for differentiated products in multiple dimensions, provid-

ing a more nuanced understanding of consumer choices and market equilibrium [Berry et al., 1995].
Overall, the BLP model, through the random-coefficients logit model and its extensions, offers a robust framework for estimating demand for differentiated products, considering consumer heterogeneity, and capturing unobserved product characteristics, thus providing valuable insights for businesses and policymakers in various industries.

3.2 Plain Logit and Nested Model

Theory Of Plain Logit
As we can see from [PyBLP Documentation, Year] we start with the plain logit model under independence of irrelevant alternatives (IIA). In this model (indirect) utility is given by

$$U_{ijt} = \alpha p_{jt} + x_{jt}\beta^{ex} + \xi_{jt} + \epsilon_{ijt}$$

where ϵ_{ijt} is distributed IID with the Type I Extreme Value (Gumbel) distribution. It is common to normalize the mean utility of the outside good to zero so that $U_{i0t} = \epsilon_{i0t}$. This gives us aggregate market shares

$$s_{jt} = \frac{\exp(\alpha p_{jt} + x_{jt}\beta^{ex} + \xi_{jt})}{1 + \sum_k \exp(\alpha p_{kt} + x_{kt}\beta^{ex} + \xi_{kt})}$$

If we take logs we get

$$\log s_{jt} = \alpha p_{jt} + x_{jt}\beta^{ex} + \xi_{jt} - \log \sum_k \exp(\alpha p_{kt} + x_{kt}\beta^{ex} + \xi_{kt})$$

and

$$\log s_{0t} = -\log \sum_k \exp(\alpha p_{kt} + x_{kt}\beta^{ex} + \xi_{kt})$$

By differencing the above we get a linear estimating equation:

$$\log s_{jt} - \log s_{0t} = \alpha p_{jt} + x_{jt}\beta^{ex} + \xi_{jt}$$

Because the left hand side is data, we can estimate this model using linear IV GMM.

Theory of Nested Logit
We can extend the logit model to allow for correlation within a group h so that

$$U_{ijt} = \alpha p_{jt} + x_{jt}\beta^{ex} + \zeta_{jt} + \bar{\epsilon}_{h(}j)_{ti} + (1-\rho)\bar{\epsilon}_{ijt}$$

Now we require that $\epsilon_{jt_i} = \bar{\epsilon}_h(j)_{ti} + (1-\rho)\bar{\epsilon}_{jt_i}$ is distributed IID with the Type I Extreme Value (Gumbel) distribution. As $\rho \to 1$, all consumers stay within their group. As $\rho \to 0$, this collapses to the IIA logit. Note that if we wanted, we could allow ρ to differ between groups with the notation $\rho_h(j)$.

This gives us aggregate market shares as the product of two logits, the within group logit and the across group logit:

$$s_{jt} = \frac{\exp\left[\frac{V_{jt}}{1-\rho}\right]}{\exp\left[\frac{V_h(j)t}{1-\rho}\right]} \cdot \frac{\exp(V_h(j)t)}{1 + \sum_h \exp(V_h t)}$$

where $V_{jt} = \alpha p_{jt} + x_{jt}\beta_{ex} + \xi_{jt}$.

After some work we again obtain the linear estimating equation:

$$\log s_{jt} - \log s_{0t} = \alpha p_{jt} + x_{jt}\beta_{ex} + \rho \log s_{j|h(j)t} + \xi_{jt}$$

where $s_{j|h(j)t} = \frac{s_{jt}}{s_{h(j)t}}$ and $s_{h(j)t}$ is the share of group $h(j)$ in market t.

3.3 Random Coefficient Models

The random coefficient logit model and the random coefficient nested logit model are both advanced discrete choice models used in econometrics and transportation research. The random coefficient logit model (RC) allows for differences in tastes across individuals, providing a highly flexible model that overcomes the limitations of the standard logit
model [Steenburgh and Ainslie, 2010]. It permits a more realistic representation of substitution patterns across products compared to the simple or nested logit models [Knittel and Metaxoglou, 2008]. Additionally, the RC model is identified and allows for products with similar characteristics to be closer substitutes, which is not possible in the logit model without random coefficients [Fox et al., 2012]. On the other hand, the random coefficient nested logit model (RCNL) is proposed as a comparison to the nested logit model, offering a more complex but flexible alternative [Grigolon and Verboven, 2014]. The RCNL model is parsimonious, requiring only one additional parameter over the nested logit model, while allowing a more flexible correlation structure among alternatives [Bhat, 1998].
In terms of estimation, the random coefficient logit model has been extended in several directions, indicating its usefulness and adaptability in various research contexts [Sándor, 2013]. Furthermore, the RC model has been used in joint models

for mode choice and escort decisions of school trips, demonstrating its applicability in practical transportation studies. On the other hand, the random coefficient nested logit model has been evaluated in the context of mode choice for school trips, showing that its prediction potential is around twice as high as that of the nested logit model [Ermagun et al., 2014].
In summary, the random coefficient logit model and the random coefficient nested logit model both offer advanced capabilities for modeling discrete choices, with the former providing flexibility in representing individual differences in tastes and substitution patterns, and the latter offering a more complex yet parsimonious alternative to the nested logit model.

3.3.1 Random Coefficient Logit Model (RCLM)

As we can see from the documentation of PyBLP, the random coefficients model extends the plain logit model by allowing for correlated tastes for different product characteristics. In this model (indirect) utility is given by:

$$u_{ijt} = \alpha_i p_{jt} + x_{jt}\beta_i^{ex} + \zeta_{jt} + \varepsilon_{ijt}$$

The main addition is that $\beta_i = (\alpha_i, \beta_{exi})$ has individual-specific subscripts i.

Conditional on β_i, the individual market share follows the same logit form as before. However, we must now integrate over heterogeneous individuals to obtain the aggregate market share:

$$s_{jt}(\alpha, \beta, \theta) = \int \frac{\exp(\alpha_i p_{jt} + x_{jt}\beta_i^{ex} + \xi_{jt})}{1 + \sum_k \exp(\alpha_i p_j t + x_k t \beta_i^{ex} + \xi_{kt})} f(\alpha_i, \beta_i|\theta) d\alpha_i d\beta_{exi}$$

Given a guess of θ, we can solve the system of nonlinear equations for the vector δ, which equates observed and predicted market share

$$s_{jt} = s_{jt}(\delta, \theta)$$

Now we can perform a linear IV GMM regression of the form

$$\delta_{jt}(\theta) = \alpha p_{jt} + x_{jt}\beta^{ex} + \xi_{jt}$$

The moments are constructed by interacting the predicted residuals $\hat{\xi}_{jt}(\theta)$ with instruments z_{jt} to form

$$\bar{g}(\theta) = \frac{1}{N} \sum_{j,t} z_{jt} \hat{\xi}_{jt}(\theta)$$

3.3.2 Random Coefficient Nested Logit Model

According to the documentation of PyBLP, incorporating parameters that measure within nesting group correlation gives the random coefficients nested logit (RCNL) model of [Brenkers and Verboven, 2006] and
[Grigolon and Verboven, 2014]. There are $h = 1, 2, \ldots, H$ nesting groups, and each product j is assigned to a group $h(j)$. The set $J_{h_t} \subseteq J_t$ denotes the products in group h and market t.

In the RCNL model, idiosyncratic preferences are partitioned into

$$\epsilon_{ijt} = \bar{\epsilon}_{ih(j)t} + (1 - \rho_{h(j)})\bar{\epsilon}_{ijt}$$

where $\bar{\epsilon}_{ijt}$ is Type I Extreme Value, and $\bar{\epsilon}_{ih(j)t}$ is distributed such that ϵ_{ijt} is still Type I Extreme Value.

The random coefficient nested logit (RCNL) model extends the traditional nested logit model, introducing the capability to accommodate random variation in consumer preferences across products. This extension offers a more flexible correlation structure among alternatives while maintaining simplicity by necessitating only a single additional parameter compared to the nested logit model [Grigolon and Verboven, 2014]. This flexibility allows the RCNL model to relax the independence of irrelevant alternatives (IIA) assumption to a greater extent, rendering it a valuable tool for modeling discrete choices in various contexts, such as transportation and economics.
The RCNL model's capacity to handle a more versatile correlation structure among alternatives proves advantageous in capturing the intricate decision-making processes of individuals, particularly in scenarios where the IIA assumption of the nested logit model may not hold. This flexibility permits the modeling of correlation between alternatives within the same nest [Bhat, 1998]. Additionally, the RCNL model's parsimonious nature, characterized by the introduction of just one additional parameter compared to the nested logit model, makes it an attractive choice for researchers aiming to strike a balance between model complexity and flexibility.
The RCNL model is particularly valuable in capturing the richness of consumer heterogeneity, as highlighted by [Gayle and Lin, 2022]. By allowing for random taste parameters, the model can effectively capture the variation in consumer preferences based on demographic characteristics, market-specific attributes, and unobserved preference shocks. This enables a more nuanced understanding of consumer behavior and the factors driving product choices.

3.4 Instrumental Variables And Fixed Effects

Instrumental variables play a crucial role in addressing endogeneity in the estimation of random coefficient logit models. [Berry et al., 1995],proposed an approximation to the optimal instruments for the random coefficient logit model, highlighting the importance of addressing endogeneity in demand estimation.
Additionally, the use of traditional instrumental variables techniques has been emphasized in the estimation of discrete-choice models of product differentiation, demonstrating the significance of instrumental variables in addressing endogeneity in demand models.
Furthermore, fixed effects are utilized to capture unobserved heterogeneity and account for the impact of unobserved product characteristics on demand. [Nevo, 1998] highlighted the use of brand fixed-effects in retrieving taste coefficients, emphasizing the role of fixed effects in controlling for unobserved factors in demand estimation. Moreover, the comparison of alternative discrete choice models of product differentiation has underscored the use of fixed effects in addressing unobserved heterogeneity in demand models.
In the context of the random coefficient nested logit model, the role of instrumental variables and fixed effects is also significant.[Berry et al., 1995] provided a flexible random coefficients logit model that accounts for the endogeneity of prices, demonstrating the importance of addressing endogeneity through appropriate modeling techniques. Additionally, the comparison of alternative discrete choice models has highlighted the use of nested logit, instrumental variables, and fixed effects in analyzing consumer choices and market competition.
In summary, instrumental variables and fixed effects are integral components of the random coefficient logit and random coefficient nested logit models. These elements are essential for addressing endogeneity, capturing unobserved heterogeneity, and improving the accuracy of demand estimation in the analysis of consumer behavior and product differentiation.

4 Data Collection

This section discusses the selection of variables critical to our research, which enables the estimation of demand elasticities and market shares for six leading cosmetic brands: Nivea, La Roche-Posay, L'Oreal, CeraVe, Neutrogena, and Garnier.

The data collection for this study was conducted in November 2023, and the data we have at our disposal is of a cross-sectional nature. This type of data collection involves gathering information over a specific time frame, providing a snapshot of that moment in time.

To compile the primary dataset necessary for our analysis, we employed web scraping techniques, specifically utilizing Selenium, to collect data from both Amazon UK and Amazon DE e-commerce platforms. We gathered the following key information for each product:

Product Name: The name of each cosmetic product.

ASIN Number: A unique identifier known as the Amazon Standard Identification Number.

Current Price: The current market price of the product.

Previous Month Sales: The sales volume from the preceding month.

Rating: Customer satisfaction ratings based on a scale.

Reviews: Qualitative feedback provided by customers.

4.1 Additional Observed Characteristics

In addition to the core dataset, we collected supplementary observed characteristics to deepen our analysis and facilitate the application of the plain Ligit, Nested Logit, Random Coefficient Logit and Random Coefficient Nested Model. These characteristics included:

Skin Type : Classification of products based on their suitability for specific skin types, such as "All Skin Types," "Combination," "Dry," "Normal," "Oily," and "Sensitive."

Time of Application: Identification of when the product is intended to be used, differentiating between all-day, daytime, and nighttime use.

Product Form: Description of the physical form of the product, including categories like "Gel," "Cream," "Serum," and "Oil."

Scented or Unscented: Notation of whether the product had a fragrance or was unscented.

It's worth noting that we had a total of 88 products that were common to both Amazon DE and Amazon UK, forming the basis of our data collection for this study.

5 Sentiment Analysis on Reviews

In this section, we delve into the comprehensive sentiment analysis conducted on the collected customer reviews. The primary objective was to augment our dataset with an additional variable: the sentiment score. These scores, typically ranging from -1 to +1, are instrumental in characterizing each review as either positive, negative, or neutral through the application of natural language processing techniques. The motivation behind this analysis was to gain a deeper understanding of how each of the most renowned cosmetic products influenced customer sentiment, shedding light on whether these influences skewed towards the positive or negative spectrum.

5.1 Rationale For Sentiment Analysis

The motivation behind incorporating sentiment analysis into our study was twofold. Firstly, we aimed to leverage the rich repository of customer reviews, particularly focusing on the most recent hundred reviews for each product, to extract valuable insights. These insights would not only provide a nuanced understanding of customer perceptions but also serve as a valuable addition to our predictive modeling endeavors.
Secondly, by quantifying the sentiment associated with each review, we sought to investigate the influence of these widely recognized cosmetic brands: Nivea, La Roche-Posay, L'Oreal, CeraVe, Neutrogena, and Garnier—on the emotional experiences of their customers. The sentiment scores were anticipated to reveal whether customers generally expressed positivity or negativity towards these products. This information could offer crucial market intelligence, aiding in strategic decision-making for product development and marketing strategies.

5.2 Data Processing For Sentiment Analysis

Translation for Language Consistency
To ensure uniformity in our analysis, we initially addressed language discrepancies within the reviews. Since the sentiment analysis tool we employed, TextBlob, primarily operates in English, we utilized the DeepL Translator API for language translation. This step ensured that all reviews were in English, facilitating a consistent and coherent sentiment analysis.
Preprocessing for Text Cleanliness
Prior to sentiment analysis, we undertook extensive data preprocessing to ensure the quality and consistency of the textual data. This preprocessing encompassed the following steps:

Character and Punctuation Removal: Special characters and punctuation were systematically removed from the text, while extra white spaces were eliminated to maintain textual cleanliness.

Text Lowercasing: To achieve uniformity and minimize redundancy, all text was converted to lowercase. Tokenization: The text was tokenized into individual words or tokens, facilitating subsequent analysis.

Stopword Removal: Common English stopwords were systematically removed from the text to reduce noise and enhance the relevance of the analysis.

5.3 Sentiment Analyis Using TEXTBLOB

For the core sentiment analysis, we leveraged the TextBlob library, which provided a robust foundation for sentiment scoring. The sentiment analysis function determined the sentiment polarity of each review, yielding a sentiment score that ranged from -1 (indicating a strongly negative sentiment) to +1 (indicating a strongly positive sentiment). Reviews with scores close to 0 were considered neutral in sentiment.
Incorporating both sentiment labels (Positive, Negative, or Neutral) and sentiment scores into our dataset allowed for a nuanced evaluation of customer sentiment associated with each product.
The sentiment analysis results, including sentiment labels and scores, were added into our dataset. These additional variables enhanced our analytical toolkit, empowering us to explore how customer sentiment interacted with various product attributes and affected market dynamics.
This comprehensive sentiment analysis not only provided deeper insights into customer perceptions but also offered valuable strategic guidance for product manufacturers and marketers. It enabled us to discern whether customer sentiment towards these renowned cosmetic products was predominantly positive or negative, shedding light on potential areas for improvement and optimization. As we can see from Figure 1, this comparative analysis draws the subtleties in customer sentiment between the UK (GB) and German (DE) markets toward similar beauty brands and products. The mean sentiment scores are brought to life across each brand in each market in the grouped bar chart.
Although the two sets of consumers are offered the same products, from the data above, one could identify slight differences in the sensibilities and feelings of the customers who purchase them across the regions. These differentiations, though not so much perceptible, are visibly seen by the scores as depicted in the plot.
This observation highlights the high significance of localized key market dynamics and customer preferences. Even in the markets which are quite similar, for exam-

ple from the UK to Germany, there is a sharp variation in customer sentiment. Such insights are valuable in the context of strategic decision-making, indicating the importance of aligning marketing approaches and product positioning with the particular sensibilities that operate within each market.

6 Methodology

6.1 Data Preprocessing

To apply the BLP model, I had to rename and organize variables in my dataset for better clarity. Here are the key variables I created or renamed:
'market_ids': IDs that associate products with markets.
'shares': Market shares.
'prices': Product prices.
'firm_ids': IDs that associate products with firms.
Additionally, I used demand-side instruments to better understand the BLP model. These instruments consist of:
'demand_instruments' (numeric): Excluded demand-side instruments.
$\mathbf{X}_{ex1}$: Formulated exogenous demand-side linear product characteristics.
$\mathbf{Z}_D$: The full set of demand-side instruments.
A critical step in preparing our dataset for the econometric analysis was the treatment of categorical variables. Our dataset comprises several categorical variables, including 'skin_type', 'application_time', 'cream_type', and 'scent_type'. These variables represent distinct categories within each feature, such as different skin types (e.g., oily, dry, combination) or cream types (e.g., gel, cream, serum). Given the categorical nature of these variables, a direct inclusion in the regression analysis would be inappropriate due to their non-numeric nature and the arbitrary assignment of numerical values that could imply an unintended ordinal relationship.
In the context of the plain logit demand model, each product is considered an independent alternative, and the choice probabilities are modeled based on the utility derived from the product characteristics, inclusive of the one-hot encoded categorical variables. These variables, transformed into binary indicators, serve to distinct the effects of each category on consumer utility and, consequently, on product demand. For instance, the presence or absence of a specific 'creamtype' (e.g., gel, cream) in a product is directly factored into the utility function, allowing us to quantify how these attributes influence consumer choice in the market.

6.2 Estimation of Market Shares

6.2.1 Product-Level Market Shares

A fundamental component of our analysis involved the computation of market shares for each product within its respective market. Market shares, representing the proportion of total sales accounted for by each product, serve as a proxy for product demand in the BLP model framework.

The calculation started with the aggregation of monthly sales data at the market level to determine the total sales volume in each market. Following this, we calculated the market share for each product by determining its sales as a percentage of the total market sales. This calculation involved dividing each product's monthly sales by the total sales of its corresponding market, thereby yielding the market share for each product.
To validate the accuracy of our market share calculations, we conducted a summation check across all products within each market. Theoretically, the sum of all market shares in a given market should closely approximate 1, reflecting the full allocation of market sales among the constituent products.
The process involved grouping products by their associated brands within each market and summing their individual market shares. This summative approach enabled us to capture the collective market presence of each brand, offering a lens through which to assess brand performance and market strategy.
The resulting brand-level market share analysis illuminated the competitive dynamics within the skincare market, revealing both dominant players and emerging contenders. It also facilitated a nuanced understanding of market segmentation and brand differentiation, which are crucial for formulating effective market entry and expansion strategies.

6.2.2 Market-Level Market Shares

Building on the product-level analysis, we extended our examination to compute market shares at the brand level. This aggregation provided insights into the competitive positioning and market influence of different brands within the skincare industry.
The process involved grouping products by their associated brands within each market and summing their individual market shares. This summative approach enabled us to capture the collective market presence of each brand, offering a lens through which to assess brand performance and market strategy.
The resulting brand-level market share analysis illuminated the competitive dynamics within the skincare market, revealing both dominant players and emerging contenders. It also facilitated a nuanced understanding of market segmentation and brand differentiation, which are crucial for formulating effective market entry and expansion strategies.
As illustrated in Figure 2, the market dynamics for skincare brands vary significantly between the United Kingdom (UK) and Germany. Notably, Cerave's market presence in the UK accounts for 0.38% of the total market share, whereas in Germany, it captures a slightly lower share of 0.22%. This discrepancy underscores the varying consumer preferences or market strategies between the two regions.
Further analysis of other brands reveals additional insights into their market per-

formance. In Germany, La Roche-Possay, L'Oréal, and Neutrogena hold market shares of 0.07%, 0.29%, and 0.27%, respectively. Conversely, in the UK, the same brands exhibit different market shares: La Roche-Posay at 0.05%, L'Oréal at 0.26%, and Neutrogena at 0.14%. This variation indicates distinct market positioning and consumer reception in each country.
In contrast to the trends observed with other brands, Nivea's market shares are closely aligned between the two countries, with the UK market share at 0.15% and the German market at 0.13%. This suggests a consistent brand performance and consumer preference for Nivea across these markets.
Lastly, Garnier's market share is notably smaller in both regions, with 0.01% in Germany and 0.02% in the UK. This minimal market share points to a more niche or constrained market presence compared to its competitors.
These findings highlight the intricate landscape of the skincare market in Europe, with each brand exhibiting unique strengths and challenges across different regions.

6.3 Creation Of Instrumental Variables

A pivotal aspect of our demand estimation process involves addressing the inherent endogeneity of prices. Endogeneity arises when prices are influenced by unobserved factors that also affect demand, leading to biased estimates. To disentangle the exogenous variation in prices attributable to differences in marginal costs from the endogenous variation linked to unobserved product valuation, we employ a two-pronged instrumental variable approach inspired by [Berry et al., 1995], herein referred to as BLP.
According to [Gandhi and Nevo, 2021] the most popular IVs are $Z_t = x_t$, namely the characteristics of all products in the market. They are informative because they can be used to measure the proximity of competition and therefore should be correlated with price and other endogenous variables. They will also be correlated with terms like the within nest share, in the Nested Logit model, or $f_j(s_t, x_t, p_t)$ in the more general model.
The question still remains which of the many possible functions $A_j(\cdot)$ of x_t should we use to construct IVs. Different suggestions have been made in the literature. [Berry et al., 1995] propose to use: (1) own product characteristics, x_{jt}, (2) sum of characteristics of the other products produced by the same firm (for multi-product firms), (3) sum of the characteristics of competitor products. The logic for this particular set is as follows. The own product characteristics are instruments "for themselves." The other two sets try to capture the logic that the price of product j (and $f_j(\cdot)$) will depend on characteristics of other products, and that the dependence differs if these are own products or competitors' products.
[Gandhi and Houde, 2020] propose to refine how we use the information in order to improve empirical performance and avoid weak IV challenges that can arise in

practice. Specifically, they use the economic structure of the model to motivate a class of IVs they term "differentiation Instruments", which intuitively capture the relative isolation of each product in characteristics space. Defining $d_{jkt} = x_{jt} - x_{kt}$ as the vector of characteristic differences between product j and product k in market t, they construct two distinct sets of differentiation IVs that are useful for applied work.

In this study, we develop "sums of characteristics" BLP (Berry, Levinsohn, and Pakes) instruments, drawing inspiration from Ghose's methodology. Our approach employs BLP-style instruments, concentrating on the observable characteristics of products and their aggregate measures across both competing and non-competing brands within a given market. This technique aims to segregate the exogenous variations in product features from the endogenous factors that affect consumer preferences, thereby enhancing the accuracy of our demand estimations. Echoing Ghose's findings, our analysis recognizes the critical need to address the potential endogeneity present in observed product characteristics. To this end, we have integrated controls for significant product updates and alterations, ensuring a rigorous examination of the data.

Traditional "sums of characteristics" BLP instruments are:
$\mathbf{Z}_{BLP}(\mathbf{X}) = [\mathbf{Z}_{BLP,Other}(\mathbf{X}), \mathbf{Z}_{BLP,Rival}(\mathbf{X})]$ where $\mathbf{X}$ denotes a matrix representing product characteristics. Within this framework, $\mathbf{Z}_{BLP,Other}(\mathbf{X})$ is a second matrix that consists of sums over characteristics of non-rival goods,and $\mathbf{Z}_{BLP,Rival}(\mathbf{X})$ is a third matrix that consists of sums over rival.

Let $\mathbf{x}jt$ represent the characteristic vector for product j in market t, produced by firm f (where $j \in Jf_t$). Consequently, the formulations for the BLP instruments are as follows:

Then, $\mathbf{Z}_{jt}^{BLP,Other}(\mathbf{X}) = \sum_{k \in J_{f_t} k \neq j} x_{kt}$, $\mathbf{Z}_{jt}^{BLP,Rival}(\mathbf{X}) = \sum_{k \in J_{f_t} k \neq j} x_{kt}$.

Subsequently, we introduce differential instruments, as delineated by the pyBLP framework and informed [Gandhi and Houde, 2017]. These instruments are articulated as:

$$\mathbf{Z}^{Diff}(\mathbf{X}) = [\mathbf{Z}^{Diff,Other}(\mathbf{X}), \mathbf{Z}^{Diff,Rival}(\mathbf{X})],$$

where $\mathbf{X}$ is the matrix of product characteristics, with $\mathbf{Z}^{Diff,Other}(\mathbf{X})$ and $\mathbf{Z}^{Diff,Rival}(\mathbf{X})$ representing matrices of sums over functions of differences for non-rival and rival products, respectively. Initially, we constructed "local" differentiation instruments, primarily comprising counts of proximate rival and non-rival products in each mar-

ket, while excluding the constant column to prevent collinearity in the differentiation instrument columns. Following this, we developed the "quadratic" version of these instruments, offering a more continuous measure through the summation of squared differences between both rival and non-rival products within each market.

6.4 Application of the Logit model

Following the development of instrumental variables, we embarked on constructing our initial logit model, which was predicated solely on the variable of price. We formulated three distinct logit models, each incorporating one of the previously devised instrumental variables: the sums of characteristics, local differentiation, and quadratic differentiation instruments. In the context of logit modeling, a simplified problem can be established by omitting the non-linear parameters (X_2) and any information pertaining to agents. This simplification necessitates only the definition of the linear demand component, which we applied to each set of variables we generated. Subsequent to the model construction, we engaged in post-estimation analysis to discern the elasticities and diversion ratios, compute the marginal costs and markups, and evaluate the potential implications of mergers among the firms. This comprehensive analysis was conducted for each of the logit models based on the singular variable of price. Extending our analysis, we then incorporated additional product attributes into the logit models, including "skin type," "application time," "cream type," "scent," "ratings," and "sentiment score." This expansion aimed to capture a broader spectrum of factors influencing consumer preferences and market dynamics. Following the integration of these additional product characteristics, we reiterated the post-estimation procedures to calculate the elasticities, diversion ratios, marginal costs, markups, and to assess the potential mergers. This two-stage methodology, initially concentrating exclusively on price before expanding to include a diverse set of product characteristics, facilitated a nuanced exploration of market dynamics. It shed light on the multifaceted factors that influence consumer choices and the competitive landscape within the sector.

7 Empirical Results

This chapter presents the empirical findings of our study, where we rigorously apply the logit models developed in the previous sections to real-world market data. Our analysis is structured to systematically unveil the impacts of price and product characteristics on consumer preferences and market competition. The chapter is organized to discuss the results obtained from the three logit models using the instrumental variables of sum of characteristics and the differantiated instrumental variables such as the local and quadradic IVs. The three models which are based on the price as the explanatory variable and also the product features, offering insights into how variables such as "skin type," "application time," "cream type," "scent," "ratings," and "sentiment score". We then delve into the post estimation of the three models where we estimated the cross-price and own-price elasticities and also a virtual merge of the brands of NIVEA and CeraVe.

7.1 Results From the Logit Models

As illustrated in Table 1, the models consistently demonstrate a negative price coefficient, affirming the fundamental economic principle of an inverse relationship between price and demand. The significance of this relationship is supported by relatively small standard errors in relation to the beta estimates, indicating that consumers exhibit price sensitivity. However, the extent of this sensitivity varies with the instrument used, with the Quad IV's model presenting a larger standard error, potentially indicating a less reliable estimation of price elasticity.

Regarding product ratings, positive coefficients across all models suggest that higher ratings bolster product demand. This aligns with the expectation that consumers are more inclined to purchase highly-rated products. Despite this positive association, the magnitude of the coefficients is not substantially large, and the large standard errors suggest caution in firmly concluding the statistical significance of ratings on demand.

The sentiment score's impact on demand displays considerable variability across the models, shifting from negative to highly positive. Coupled with large standard errors, these inconsistent estimates indicate that sentiment scores may not be a robust predictor of demand within this context.

Focusing on the attributes of product characteristics, which are represented by dummy variables, it is important to acknowledge that one level is omitted in each category to serve as the reference point. For skin type, 'Sensitive' is the omitted category. Most coefficients within the skin type categories are positive, except for 'Normal' in the first model, which suggests a general consumer preference for other skin types over 'Sensitive'. However, only some of these coefficients are statisti-

cally significant, suggesting that while there is a preference trend, it may not be strong.
In the application time category, 'Day' is the omitted level. The positive coefficients for 'All Day' and 'Night' suggest that these application times are preferred over 'Day'. The 'Night' application time is statistically significant in the first two models, reinforcing its positive impact on demand.
For the cream type category, 'Serum' is the omitted level. Positive coefficients for 'Cream' and 'Gel' indicate a preference for these types over 'Serum', with 'Gel' showing a strong preference, particularly in the Quad IV's model. Conversely, 'Oil' is less preferred, as indicated by negative coefficients in two of the models.
Lastly, the scent category, with 'Unscented' as the omitted level, consistently presents negative coefficients for 'Scented' across all models. This suggests a general consumer preference for 'Unscented' products.

7.2 Post-Estimation

The first step in our post-estimation was to estimate the elasticities and diversion ratios of the three models. Next we moved to the case were a hypotheticall merger betwwen NIVEA and CeraVe could happen and the effects that could have to the prices of the products.

7.2.1 Elasticities

First we starting by computing the elasticities of the two markets. As we can see from the *figure 3, 4 and 5*, the results are just like we expectesd for all of the three modeles.
Economic theory posits that the diagonal elements of the elasticity matrix should represent own-price elasticities, which are expected to be negative, indicating that an increase in the price of a product leads to a decrease in its demand. This theoretical expectation is confirmed in our analysis, as the diagonal elements in all three plots are indeed negative.
Furthermore, the off-diagonal elements, representing cross-price elasticities, are indeed positive and exhibit relatively lower magnitudes compared to the own-price elasticities. This suggests that the products in question are substitutes to some degree, as an increase in the price of one product leads to an increase in the demand for others, albeit the effect is less pronounced than the impact of own-price changes.
From the above plots, we extracted the cross-price elasticities across the UK and DE markets, as presented in Tables 3, 4, and 5, reveals the nature of relationships between different beauty products within each market.
In our examination of the Sum IV model as we can see in Table 3, we observe clear

patterns of substitutability among UK product pairs, such as UK_Product_24 with UK_Products_18, 66, and 65. These products exhibit positive cross-price elasticities, albeit with moderate magnitudes, suggesting a level of substitutability that, while significant, is not pronounced. The consistency in elasticity values among these pairs positions UK_Product_24 as a potentially pivotal substitute within the UK market landscape, indicating its centrality in consumer choice as an alternative to several other products.

Conversely, the DE market presents a scenario of more subdued substitutability, as reflected by the lower elasticities among DE product pairs. This suggests a market characterized by higher product differentiation or consumer loyalty, where price changes in one product are less likely to drive significant shifts in demand towards others.

The analysis of complementary pairs within the Sum IV model reveals an intriguing contrast. UK products display positive elasticities that hover near zero, denoting a very weak or negligible level of complementarity. This finding suggests that, within the UK market, the interdependencies between products are minimal, with price changes in one product having little to no impact on the demand for another. In stark contrast, the DE_Product_1 and DE_Product_87 pair stands out with a marked negative elasticity, indicative of a strong complementary relationship. In this case, a price increase in one product leads to a notable decrease in demand for the other, highlighting a significant interdependence between these two products in the DE market.

The Local IV model (Table 4) largely corroborates the Sum IV model's findings regarding substitutable pairs, with UK_Product_24 retaining its substitutive role, albeit with a slightly attenuated strength. This consistency across models reinforces the notion of UK_Product_24's substitutive importance. The DE market continues to exhibit limited substitutability, aligning with previous observations. For complementary pairs, the UK market's results remain consistent with the Sum IV model, displaying weak positive elasticities. The DE_Product_1 and DE_Product_87 pair, however, while still demonstrating a complementary relationship, show a reduced negative elasticity, suggesting a somewhat diminished but still significant interdependence.

Transitioning to the Quad IV model (Table 5), we notice an intensification in the substitutability dynamics involving U_Product_24, as evidenced by increased positive elasticities with UK_Products_23, 69, and 5. This enhancement suggests a more robust substitutive relationship than observed in earlier models, pointing to a possibly greater responsiveness of consumers to price changes among these products. In the DE market, a slight uptick in elasticity values hints at an incremental increase in substitutability, though it remains relatively subdued. The complementary pairs in the UK market persist in showing weak positive elasticities, main-

taining the trend of minimal interdependence. Remarkably, the DE_Product_1 and DE_Product_88 pair exhibit an even more pronounced negative elasticity than in previous models, underscoring a strengthening of their complementary relationship.

Synthesizing the insights from all models, UK_Product_24 emerges as a key substitutive product within the UK market, suggesting its role as a central alternative for consumers. Conversely, the DE_Product_1 and DE_Product_88 pair manifest as markedly complementary within the DE market, indicating a strong mutual dependence that is accentuated across different modeling approaches. The variation in the strength of these relationships across models underscores the influence of model selection on the interpretation of market dynamics. It is imperative to consider both the statistical and economic significance of elasticity magnitudes, particularly concerning complementary goods, to fully grasp the nuances of product interdependencies. This analysis not only sheds light on the complex fabric of market interactions but also underscores the need for a nuanced approach to model selection and interpretation in the context of market analysis.

7.2.2 Diversion Ratios

Diversion ratios serve as a critical lens through which to examine the competitive dynamics of a market, offering insights into how demand shifts between products when one becomes less accessible or exits the market altogether. Such shifts can be triggered by various market events, including substantial price increases that effectively reduce a product's market presence.

In Table 6, we delve into the diversion ratios for two products that command the largest market shares in their respective markets, the UK and DE. These products were selected for analysis due to their significant influence within their markets.

Within the UK market, the analysis reveals that in the hypothetical scenario of UK_Product_21's withdrawal, UK_Product_24 and UK_Product_18 emerge as the primary alternatives, each garnering a diversion ratio of 0.076877. This equivalence in diversion ratios points to a potentially balanced consumer preference between these substitutes, which may stem from similar product attributes or consumer perceptions that closely associate these products with UK_Product_21. Following closely is UK_Product_65, with a diversion ratio of 0.059793, positioning it as a viable but slightly less favored option for consumers previously opting for UK_Product_21.

Turning our attention to the DE market, the focus shifts to DE_Product_49. The analysis indicates a significant diversion of demand to DE_Product_1, evidenced by a diversion ratio of 0.0100793, with DE_Product_86 trailing slightly with a ratio of 0.0100419. Although these diversion ratios are modest, they underscore the role of these products as feasible substitutes for DE_Product_49's consumer base.

DE_Product_77, with a lower diversion ratio of 0.00455277, is identified as a less preferred substitute, suggesting a lesser degree of consumer inclination towards this product in comparison to DE_Product_1 and DE_Product_86.
This exploration of diversion ratios not only highlights the potential market dynamics following the hypothetical removal of key products but also underscores the intricate web of consumer preferences and product substitutability that characterizes the competitive landscape of the UK and DE markets. Through this lens, we gain a nuanced understanding of the market's structure, revealing the layers of consumer choice and the potential ripple effects of market changes.

7.2.3 Merger

In the scenario of the hypothetical merger between NIVEA and CeraVe, a significant shift in the pricing dynamics for products within both the UK and DE markets is evident, as outlined in Tables 7, 8, and 9. These tables shows the pre-merger prices against those observed subsequent to the merger, providing a quantitative foundation for assessing the merger's impact on pricing strategies.
Visual representations in Figures 9, 10, 11, 12, 13, and 14 further illuminate these shifts. Across all three analytical models, a distinct trend emerges: products under the NIVEA and CeraVe umbrella exhibit a notable increase in prices post-merger. This trend suggests a strategic recalibration of pricing to leverage the enhanced market positioning afforded by the merger. The graphical analyses reveal not only the escalated pricing strategies employed by the newly merged entities but also the ripple effect this recalibration has on the broader market.
Interestingly, products from competing firms also register an uptick in prices, albeit to a more subdued extent. This nuanced pricing adjustment among competitors may signal a strategic alternation within the market, as rival firms navigate the new pricing paradigm established by the merged firms. The relatively modest price adjustments for products outside the NIVEA-CeraVe merger sphere could be indicative of a tactical maneuver. These firms, in an attempt to safeguard their market shares, might have opted for marginal price increases. This strategy serves a dual purpose: capitalizing on the general upward trend in market prices while maintaining a competitive edge as cost-effective alternatives to the offerings of the newly merged entity.
The pronounced price increases for NIVEA and CeraVe products post-merger could be attributed to the enhanced pricing power that typically accompanies such corporate consolidations. This power enables the merged entity to elevate prices with minimal risk to its market share, possibly buoyed by an augmented perception of value among consumers. Conversely, the restrained price increases for non-merged products reflect a delicate balancing act by competing firms. These entities aim to navigate the altered market landscape by incrementally increasing prices, thus

partaking in the general uplift in market pricing, yet remain vigilant to retain their competitive positioning.

This analysis not only sheds light on the immediate pricing implications of the merger but also offers insights into the broader strategics within the market. It underscores the interconnection of market dynamics, where a significant move by one entity can set off a cascade of strategic responses, reshaping the competitive terrain.

8 Conclusions

This thesis was aimed to explore the dynamic and complex nature of demand within the beauty product market on UK and DE Amazon platforms. Employing the BLP (Berry, Levinsohn, and Pakes) approach and utilizing the pyBLP package, we have estimated demand and cross-price elasticities, diversion ratios, and the potential impact of a merger between NIVEA and CeraVe. The analysis has revealed insightful informations regarding consumer preferences, price sensitivity, the substitutability between products, and the consequences of a merger.

The findings indicate a significant degree of price sensitivity among consumers, with price coefficients being negative across all models, underscoring the inverse relationship between price and demand as predicted by economic theory. The estimated diversion ratios have unveiled the competitive positioning of products, providing a window into consumer choice behavior when faced with product unavailability.

The examination of a hypothetical merger has highlighted the power such a consolidation wields, with merged products displaying a significant price increase, suggesting enhanced market power.

The variations in outcomes across different models underscore the need for further research to validate and refine these findings. Subsequent studies should aim to extend the robustness checks, explore additional sources of endogeneity, and employ alternative instruments to ensure the results' reliability.

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Table 1: Comparison of BLP Logit Model Estimates With BLP Style Instruments

Variable	Sum IV's	Local IV's	Quad IV's
Price	-0.07101	-0.04832	-0.02972
	(0.02769)	(0.02729)	(0.18614)
Ratings	0.61761	0.86028	0.85125
	(0.86537)	(0.96777)	(0.97184)
Sentiment Score	-0.90442	0.90023	5.87717
	(2.63564)	(2.46352)	(9.58410)
Skin Type: All	0.35103	0.48036	0.34389
	(0.30816)	(0.29372)	(0.30414)
Skin Type: Combination	0.46867	0.54992	0.61929
	(0.46332)	(0.49191)	(0.70093)
Skin Type: Dry	0.27002	0.99941	0.71427
	(0.34674)	(0.41750)	(0.37684)
Skin Type: Normal	-0.24867	0.00733	0.03889
	(0.46448)	(0.40621)	(0.73806)
Skin Type: Oily	0.34318	0.40365	0.22417
	(0.46114)	(0.41004)	(0.53926)
Application Time: All Day	0.19330	0.24970	0.45946
	(0.24617)	(0.25914)	(0.46754)
Application Time: Night	0.58752	0.69539	0.42237
	(0.26012)	(0.29256)	(0.51585)
Cream Type: Cream	0.02289	0.30596	1.20950
	(0.46288)	(0.42622)	(2.24314)
Cream Type: Gel	0.91673	0.63660	1.91141
	(0.55532)	(0.56130)	(2.36840)
Cream Type: Oil	-0.77861	-0.40755	1.15454
	(0.76360)	(0.71144)	(3.62388)
Scent Type: Scented	-0.22939	-0.20155	-0.27311
	(0.24815)	(0.26093)	(0.41026)

Note: The table reports the BLP logit model estimates. Robust standard errors are in parentheses below the estimates.

Table 2: Product Names

Identifier	**Product Name**
prod_1	NIVEA Hyaluron Cellular Filler Anti-Age Day Cream SPF 15 (50ml)
prod_2	NIVEA Hyaluron Cellular Filler Anti-Age Night Cream
prod_3	NIVEA Hyaluron Cellular Filler Anti-Age Day Cream
prod_4	NIVEA Light Moisturising Day Cream (50ml)
prod_5	NIVEA MEN Hyaluron Face Cream (50ml), Anti Wrinkle Cream
prod_6	NIVEA MEN Sensitive Face Moisturiser, Men's Moisturiser
prod_7	NIVEA MEN Skin Energy Moisturising Creme (50 ml), Face Cream for Men
prod_8	NIVEA Q10 Anti-Wrinkle Power Pore Minimising Day Cream SPF 15 (50ml),
prod_9	NIVEA Q10 Anti-Wrinkle Sensitive Revitalising Night Cream (50ml)
prod_10	NIVEA Q10 Energy Healthy Glow Face Day Cream
prod_11	NIVEA Q10 Energy Recharging Face Night Cream (50ml)
prod_12	NIVEA Q10 Power 60 + Skin Anti-Wrinkle + Replenishing Night Cream (50 ml)
prod_13	NIVEA Sensitive Day Cream (50 ml)
prod_14	NIVEA Sensitive Night Cream (50 ml)
prod_15	NIVEA Soft Rose 24h Day Cream (50 ml)
prod_16	Nivea Cellular Luminous 630 anti-pigment spot intensive serum (30 ml)
prod_17	Nivea Q10 Energy Anti-Wrinkle Pearl Serum (30 ml)
prod_18	CeraVe AM Facial Moisturising Lotion SPF30 with Ceramides
prod_19	CeraVe Anti Aging Retinol Serum
prod_20	CeraVe Blemish Control Gel Moisturiser with
prod_21	CeraVe Moisturising Cream for Dry to Very Dry Skin
prod_22	CeraVe Moisturising Lotion, with hyaluronic acid and 3 essential ceramides
prod_23	CeraVe PM Daily Facial Moisturiser Lotion for Normal to Dry Skin 52ml
prod_24	CeraVe Resurfacing Retinol Serum with Ceramides
prod_25	CeraVe SA Smoothing Cream for Rough and Bumpy Skin
prod_26	CeraVe Skin Renewing Gel Oil
prod_49	L'Oreal Paris Age Perfect Collagen Expert Retightening Care Night Cream
prod_65	L'Oréal Paris Revitalift Day Cream, Anti-Wrinkle Moisturiser, Day Cream SPF 30
prod_66	L'Oreal Paris Retinol & Niacinamide Night Cream
prod_69	NEUTROGENA® Clear & Radiant Face
prod_77	Neutrogena Hydro Boost Gel Cream
prod_86	Garnier SkinActive Vitamin C Face Moisturiser Brightening Day Cream, 50ml
prod_87	Garnier Youth Organic Rosy Glow 3 in1 Youth Cream with Rosehip Seed Oil, White

Table 3: Top 3 Substitute and Complementary Product Pairs Sum IV model

Product Pair	Elasticity	Market
Substitute Pairs		
UK_Product_24 & UK_Product_18	0.069862	UK
UK_Product_24 & UK_Product_66	0.069862	UK
UK_Product_24 & UK_Product_65	0.069862	UK
DE_Product_1 & DE_Product_69	0.002021	DE
DE_Product_1 & DE_Product_40	0.002021	DE
DE_Product_1 & DE_Product_28	0.002021	DE
Complementary Pairs		
UK_Product_58 & UK_Product_35	0.00012	UK
UK_Product_58 & UK_Product_40	0.00012	UK
UK_Product_58 & UK_Product_43	0.00012	UK
DE_Product_1 & DE_Product_87	-0.482333	DE
DE_Product_1 & DE_Product_31	0.002021	DE
DE_Product_1 & DE_Product_36	0.002021	DE

Table 4: Top 3 Substitute and Complementary Product Pairs Local IV model

Product Pair	Elasticity	Market
Substitute Pairs		
UK_Product_24 & UK_Product_42	0.055405	UK
UK_Product_24 & UK_Product_12	0.055405	UK
UK_Product_24 & UK_Product_66	0.055405	UK
DE_Product_1 & DE_Product_85	0.001603	DE
DE_Product_1 & DE_Product_30	0.001603	DE
DE_Product_1 & DE_Product_41	0.001603	DE
Complementary Pairs		
UK_Product_58 & UK_Product_43	0.000095	UK
UK_Product_58 & UK_Product_29	0.000095	UK
UK_Product_58 & UK_Product_26	0.000095	UK
DE_Product_1 & DE_Product_87	-0.382520	DE
DE_Product_1 & DE_Product_33	0.002021	DE
DE_Product_1 & DE_Product_18	0.001603	DE

Table 5: Top 3 Substitute and Complementary Product Pairs Quad IV model

Product Pair	Elasticity	Market
Substitute Pairs		
UK_Product_24 & UK_Product_23	0.110263	UK
UK_Product_24 & UK_Product_69	0.110263	UK
UK_Product_24 & UK_Product_5	0.110263	UK
DE_Product_1 & DE_Product_67	0.00319	DE
DE_Product_1 & DE_Product_52	0.00319	DE
DE_Product_1 & DE_Product_34	0.00319	DE
Complementary Pairs		
UK_Product_58 & UK_Product_44	0.000189	UK
UK_Product_58 & UK_Product_80	0.000189	UK
UK_Product_58 & UK_Product_29	0.000189	UK
DE_Product_1 & DE_Product_87	-0.761268	DE
DE_Product_1 & DE_Product_33	0.003190	DE
DE_Product_1 & DE_Product_18	0.003190	DE

Table 6: Top 3 Diversion Ratios For The 3 Models

Product Pair	Div. Ratios
UK_Product_21	
UK_Product_21 & UK_Product_24	0.076877
UK_Product_21 & UK_Product_18	0.076877
UK_Product_21 & UK_Product_65	0.059793
DE_Product_49	
DE_Product_49 & DE_Product_1	0.0100793
DE_Product_49 & DE_Product_86	0.0100419
DE_Product_49 & DE_Product_77	0.00455277

Table 7: Price Comparison with Sum IVS: UK & DE Post NIVEA-CeraVe Merger

UK Products	Original Price	Changed Price	DE Products	Original Price	Changed Price
prod_1	4.20	13.5149	prod_1	2.45	6.71197
prod_2	8.49	17.8049	prod_2	20.09	24.352
prod_3	8.49	17.8049	prod_3	13.95	18.212
prod_4	2.49	11.8049	prod_4	8.35	12.612
prod_5	10.95	20.2649	prod_5	13.99	18.252
prod_6	5.35	14.6649	prod_6	4.90	9.16197
prod_7	6.66	15.9749	prod_7	12.99	17.252
prod_8	10.99	20.3049	prod_8	11.95	16.212
prod_9	13.00	22.3149	prod_9	20.62	24.882
prod_10	6.87	16.1849	prod_10	19.92	24.182
prod_11	6.87	16.1849	prod_11	18.26	22.522
prod_12	8.00	17.3149	prod_12	17.61	21.872
prod_13	4.00	13.3149	prod_13	11.98	16.242
prod_14	5.00	14.3149	prod_14	14.98	19.242
prod_15	5.00	14.3149	prod_15	6.95	11.212
prod_16	14.99	24.3049	prod_16	28.95	33.212
prod_17	22.91	32.2249	prod_17	13.95	18.212
prod_18	15.00	18.2509	prod_18	14.99	17.3835
prod_19	15.99	19.2409	prod_19	38.55	40.9435
prod_20	11.60	14.8509	prod_20	12.71	15.1035
prod_21	13.60	16.8509	prod_21	17.88	20.2735
prod_22	11.00	14.2509	prod_22	12.80	15.1935
prod_23	12.40	15.6509	prod_23	13.64	16.0335
prod_24	16.19	19.4409	prod_24	16.99	19.3835
prod_25	13.80	17.0509	prod_25	14.58	16.9735
prod_26	26.11	29.3609	prod_26	40.99	43.3835

Table 8: Price Comparison with Local IVS: UK & DE Post NIVEA-CeraVe Merger

UK Products	Original Price	Changed Price	DE Products	Original Price	Changed Price
prod_1	4.20	17.89	prod_1	2.45	8.71398
prod_2	8.49	22.18	prod_2	20.09	26.354
prod_3	8.49	22.18	prod_3	13.95	20.214
prod_4	2.49	16.18	prod_4	8.35	14.614
prod_5	10.95	24.64	prod_5	13.99	20.254
prod_6	5.35	19.04	prod_6	4.9	11.164
prod_7	6.66	20.35	prod_7	12.99	19.254
prod_8	10.99	24.68	prod_8	11.95	18.214
prod_9	13.00	26.69	prod_9	20.62	26.884
prod_10	6.87	20.56	prod_10	19.92	26.184
prod_11	6.87	20.56	prod_11	18.26	24.524
prod_12	8.00	21.69	prod_12	17.61	23.874
prod_13	4.00	17.69	prod_13	11.98	18.244
prod_14	5.00	18.69	prod_14	14.98	21.244
prod_15	5.00	18.69	prod_15	6.95	13.214
prod_16	14.99	28.68	prod_16	28.95	35.214
prod_17	22.91	36.60	prod_17	13.95	20.214
prod_18	15.00	19.78	prod_18	14.99	18.5079
prod_19	15.99	20.77	prod_19	38.55	42.0679
prod_20	11.60	16.38	prod_20	12.71	16.2279
prod_21	13.60	18.38	prod_21	17.88	21.3979
prod_22	11.00	15.78	prod_22	12.8	16.3179
prod_23	12.40	17.18	prod_23	13.64	17.1579
prod_24	16.19	20.97	prod_24	16.99	20.5079
prod_25	13.80	18.58	prod_25	14.58	18.0979
prod_26	26.11	30.89	prod_26	40.99	44.5079

Table 9: Price Comparison with Quad IVS: UK & DE Post NIVEA-CeraVe Merger

UK Products	Original Price	Changed Price	DE Products	Original Price	Changed Price
prod_1	4.20	11.0791	prod_1	4.2	11.0791
prod_2	8.49	15.3691	prod_2	8.49	15.3691
prod_3	8.49	15.3691	prod_3	8.49	15.3691
prod_4	2.49	9.36913	prod_4	2.49	9.36913
prod_5	10.95	17.8291	prod_5	10.95	17.8291
prod_6	5.35	12.2291	prod_6	5.35	12.2291
prod_7	6.66	13.5391	prod_7	6.66	13.5391
prod_8	10.99	17.8691	prod_8	10.99	17.8691
prod_9	13.00	19.8791	prod_9	13	19.8791
prod_10	6.87	13.7491	prod_10	6.87	13.7491
prod_11	6.87	13.7491	prod_11	6.87	13.7491
prod_12	8.00	14.8791	prod_12	8	14.8791
prod_13	4.00	10.8791	prod_13	4	10.8791
prod_14	5.00	11.8791	prod_14	5	11.8791
prod_15	5.00	11.8791	prod_15	5	11.8791
prod_16	14.99	21.8691	prod_16	14.99	21.8691
prod_17	22.91	29.7891	prod_17	22.91	29.7891
prod_18	15.00	17.4009	prod_18	15	17.4009
prod_19	15.99	18.3909	prod_19	15.99	18.3909
prod_20	11.60	14.0009	prod_20	11.6	14.0009
prod_21	13.60	16.0009	prod_21	13.6	16.0009
prod_22	11.00	13.4009	prod_22	11	13.4009
prod_23	12.40	14.8009	prod_23	12.4	14.8009
prod_24	16.19	18.5909	prod_24	16.19	18.5909
prod_25	13.80	16.2009	prod_25	13.8	16.2009
prod_26	26.11	28.5109	prod_26	26.11	28.5109

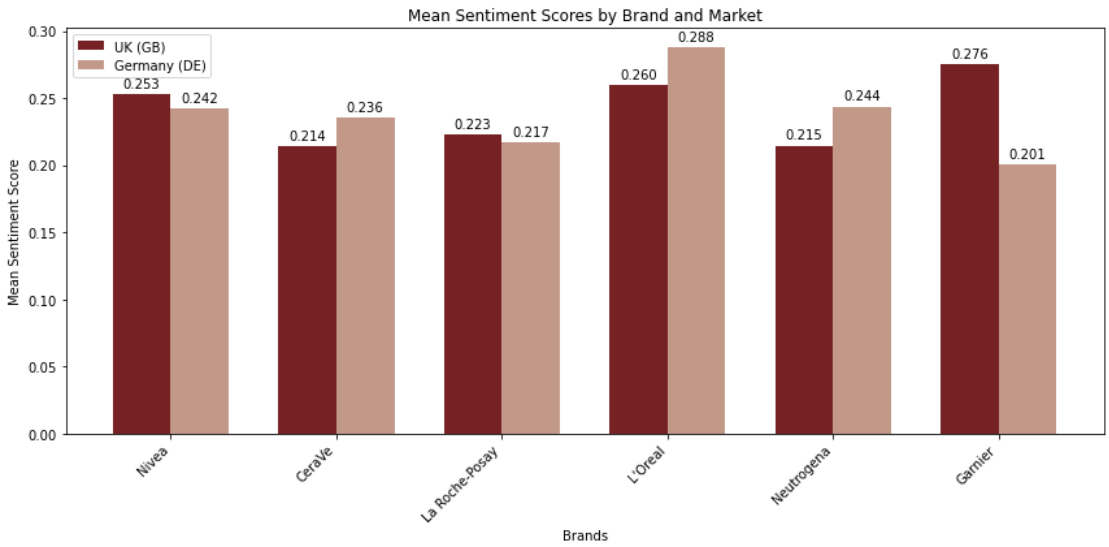


Figure 1: Sentiment Score

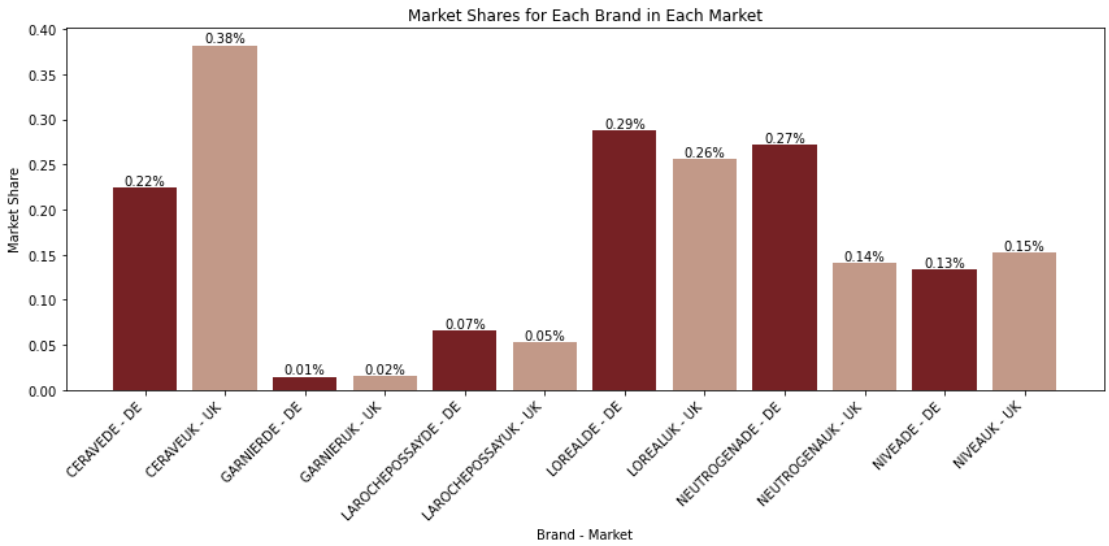


Figure 2: Market Shares

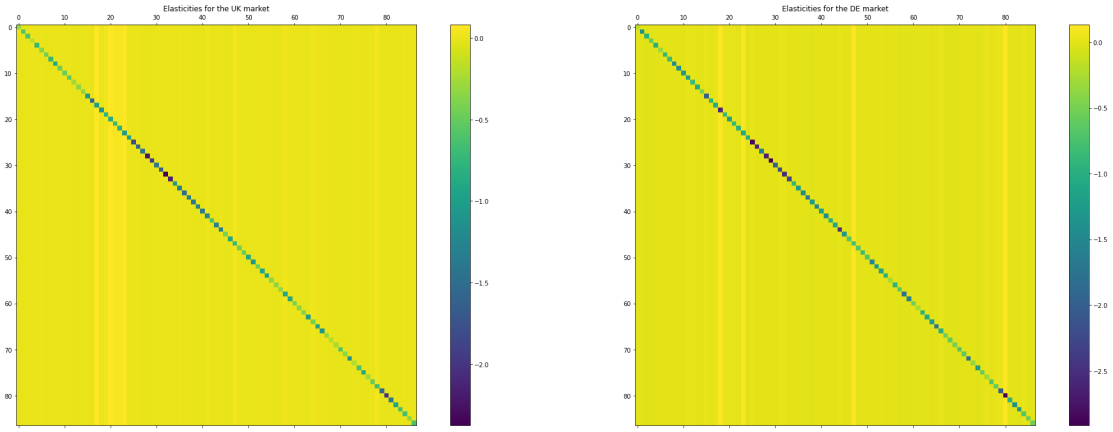


Figure 3: Elasticities for the two markets for the model with Sum IV's

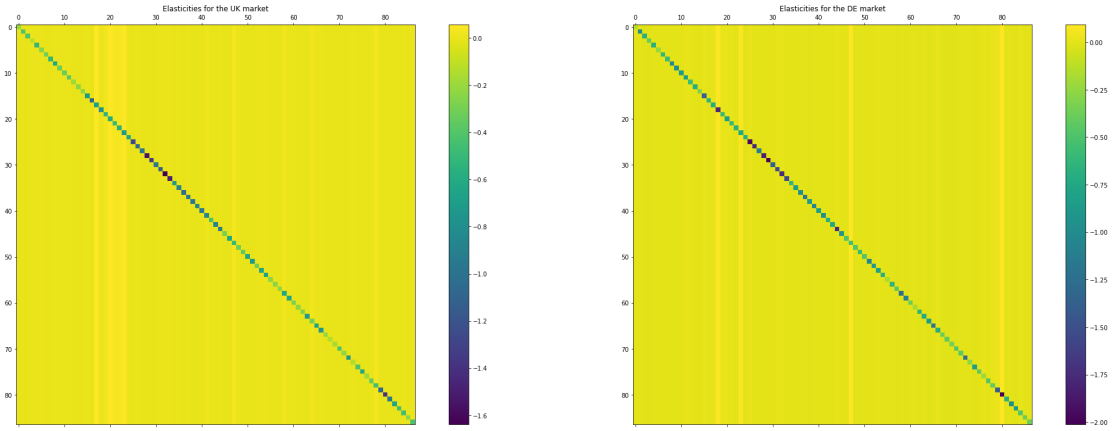


Figure 4: Elasticities for the two markets for the model with Local IV's

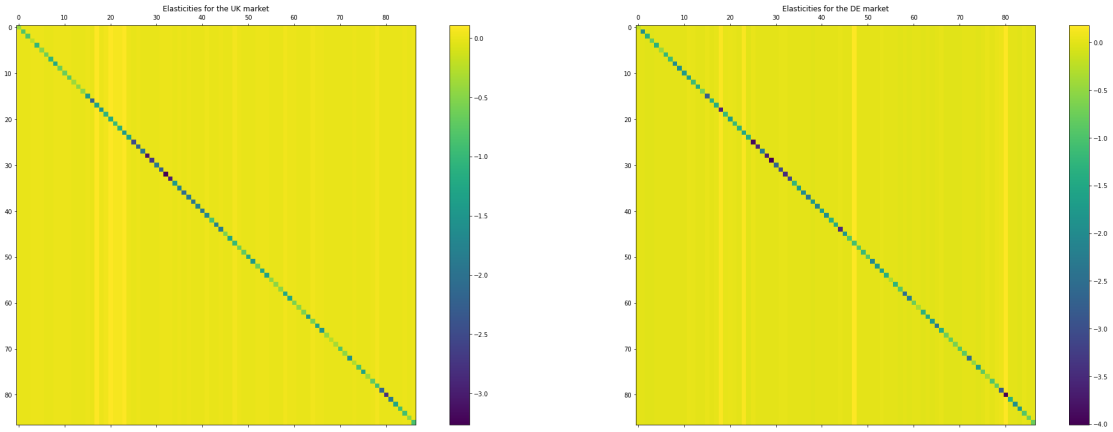


Figure 5: Elasticities for the two markets for the model with Local IV's

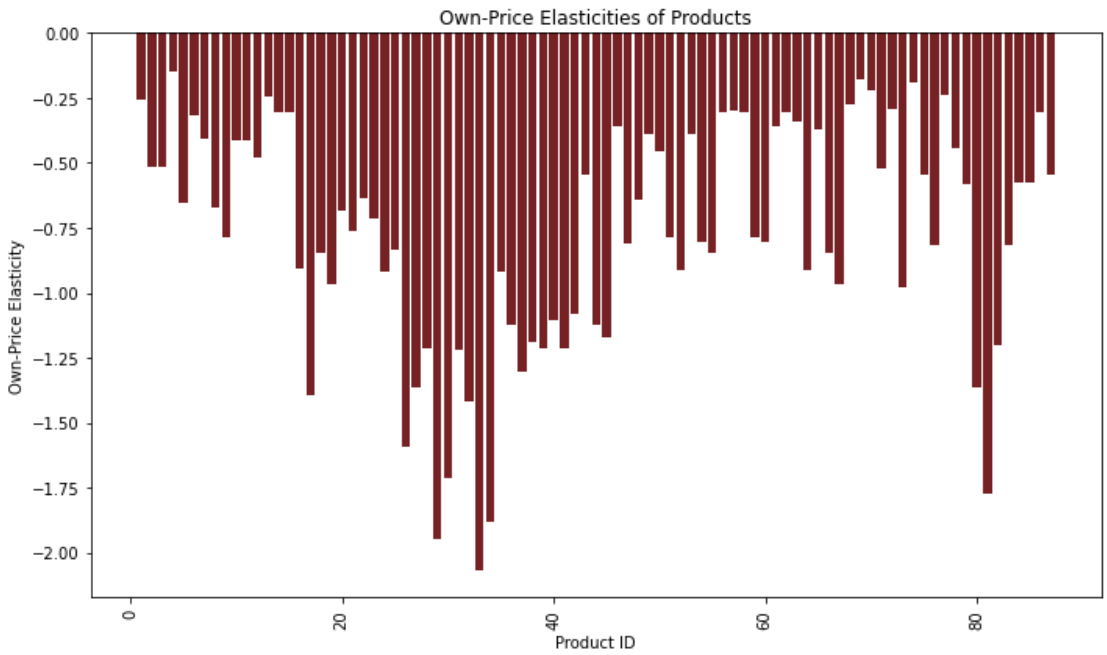


Figure 6: Own-Price Elasticities of the Sum Model

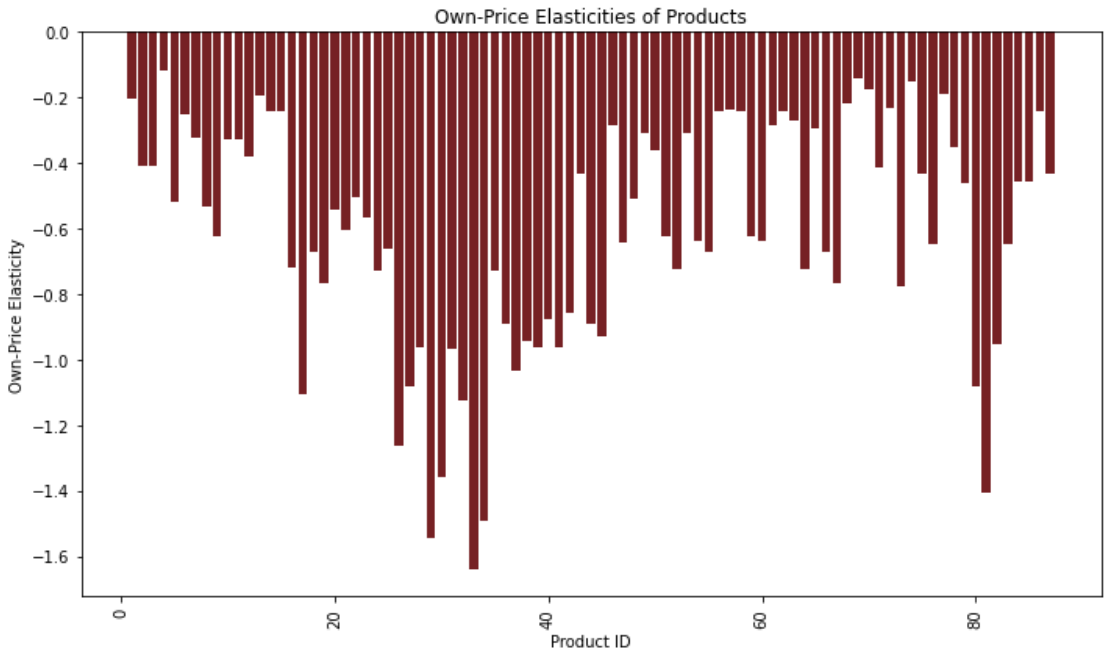


Figure 7: Own-Price Elasticities of the Local Model

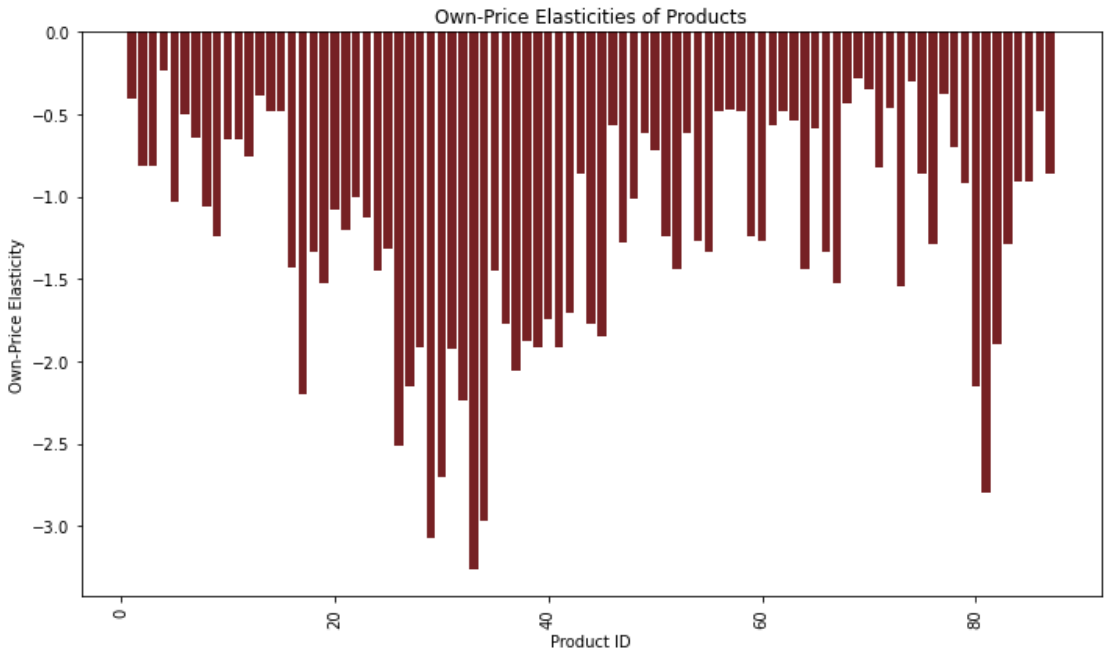
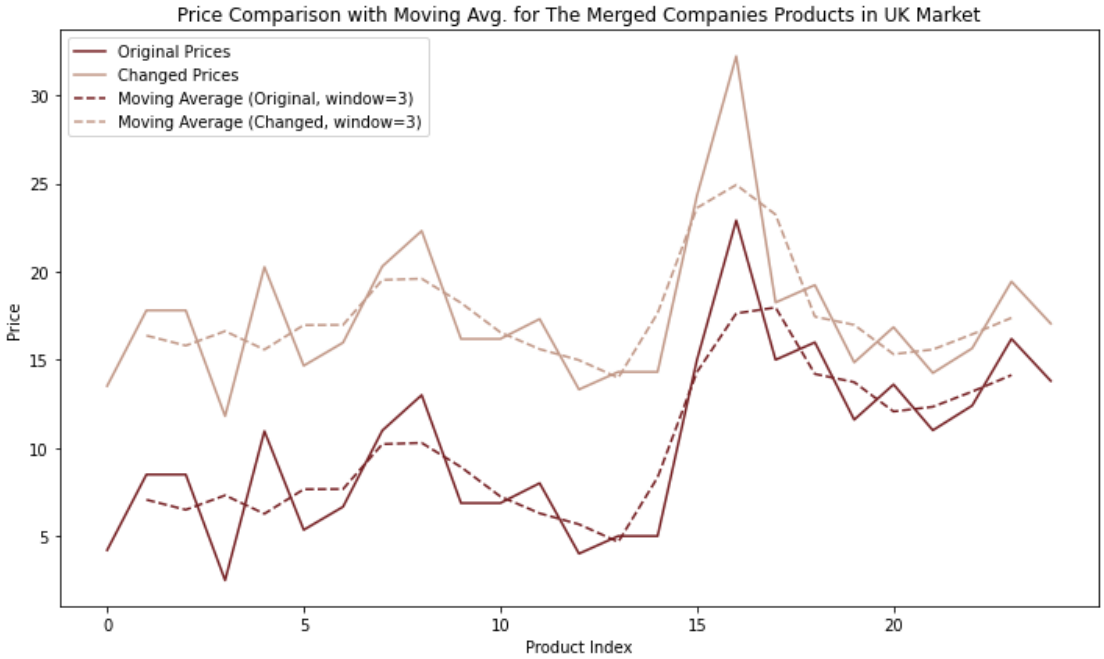


Figure 8: Own-Price Elasticities of the Quad Model



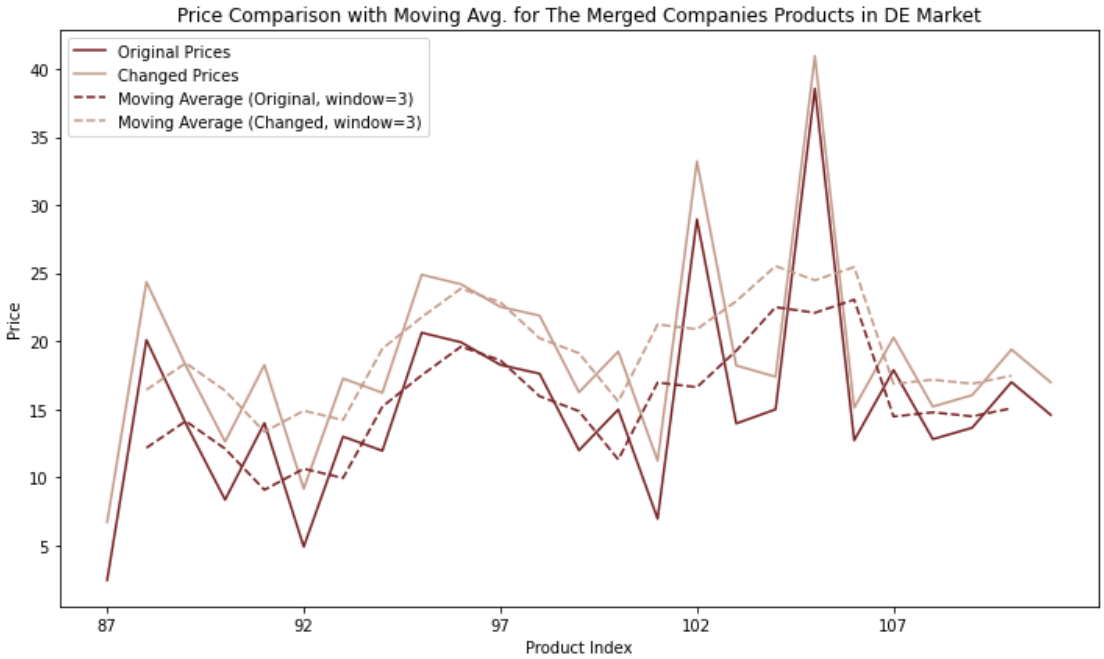
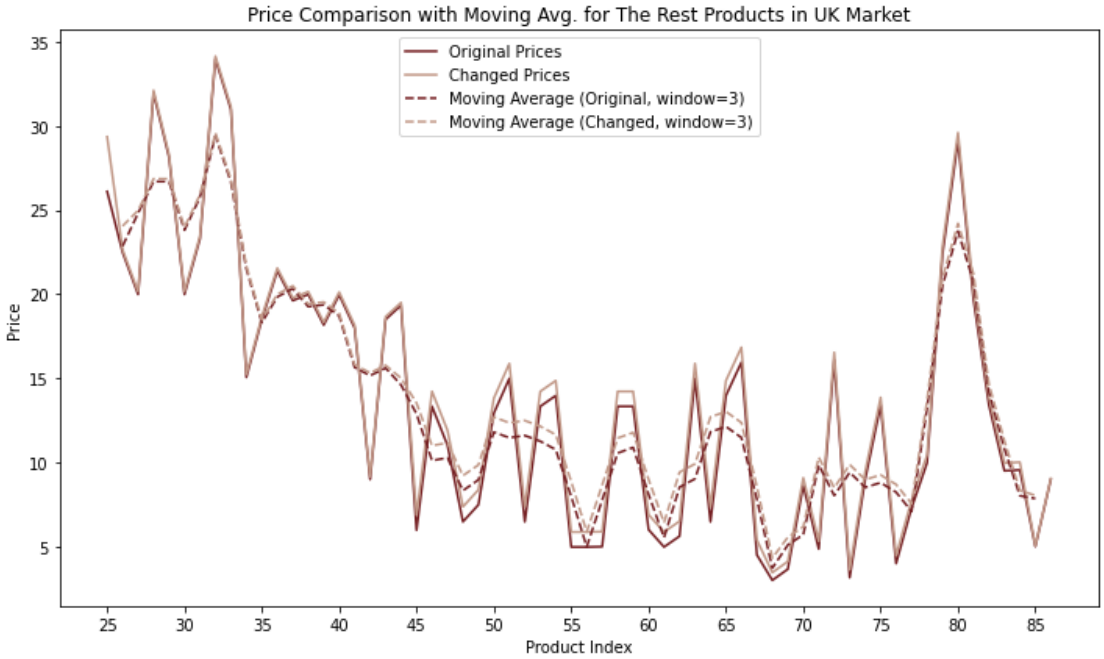


Figure 9: Price Comparison with Moving Avg. for The Merged Companies Products in UK & DE MArket (Sum IVS)



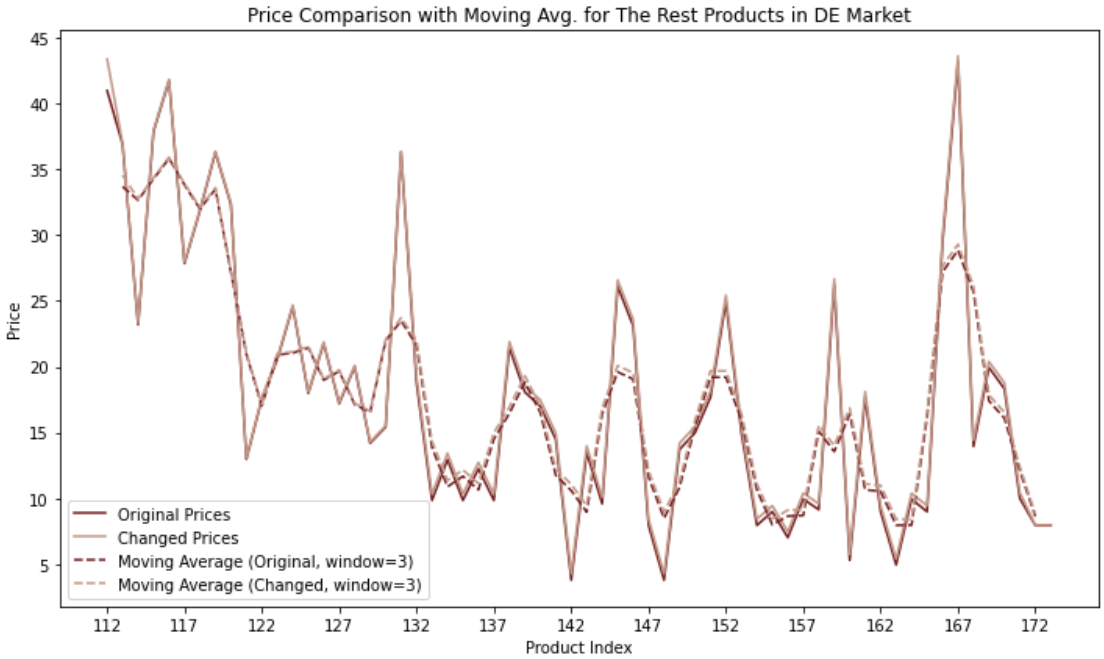
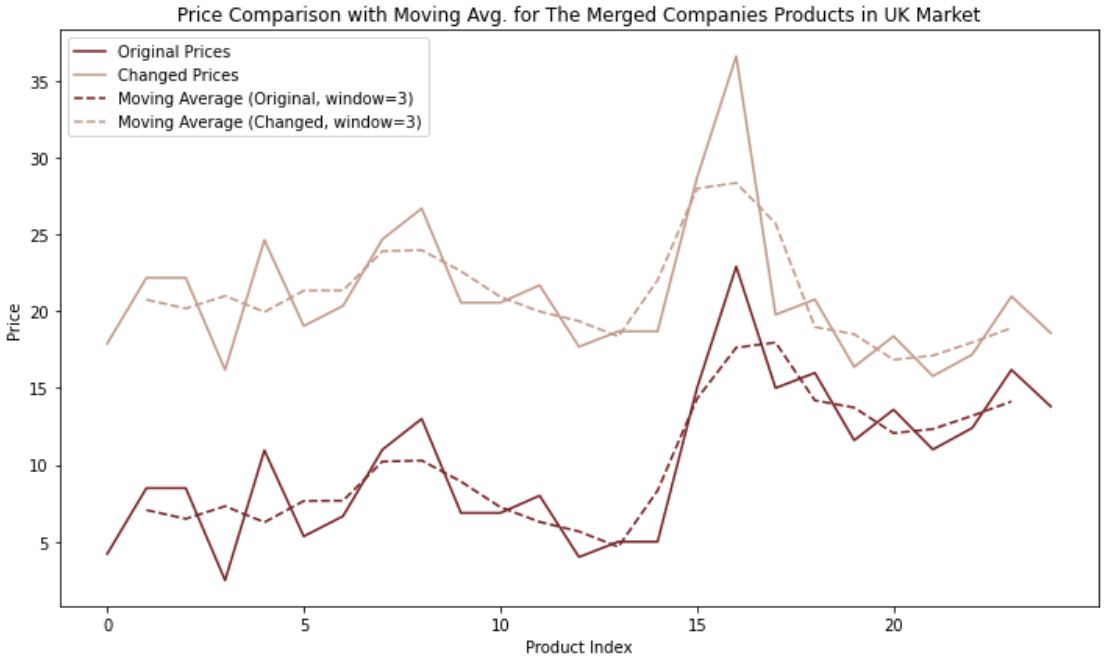


Figure 10: Price Comparison with Moving Avg. for The Rest Products in UK & DE Market (Sum IVS)



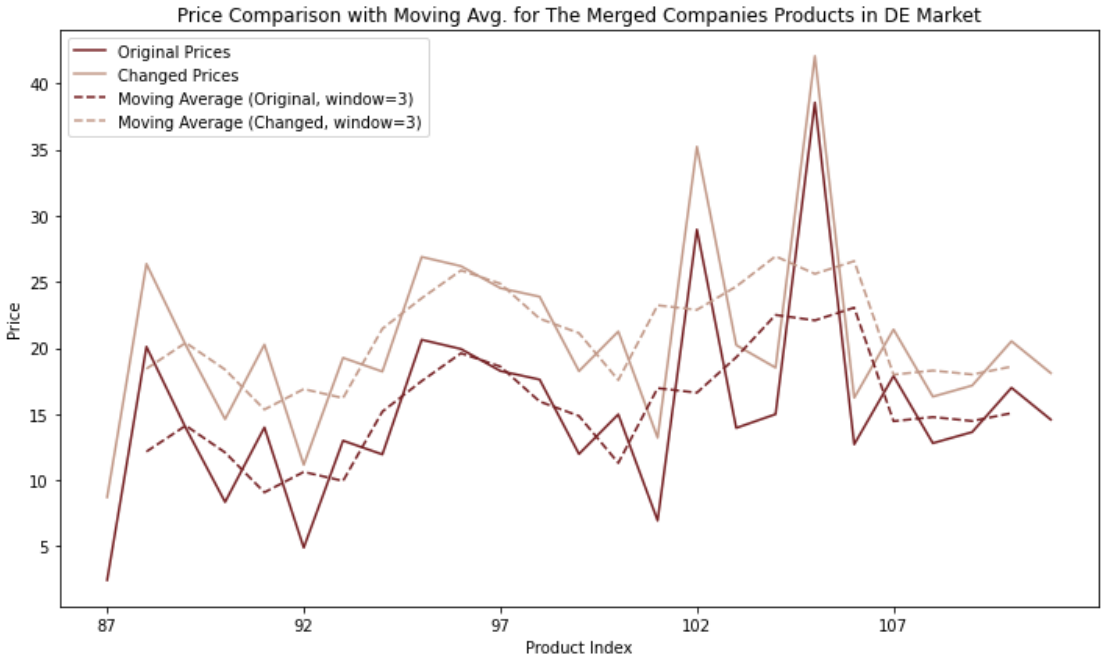
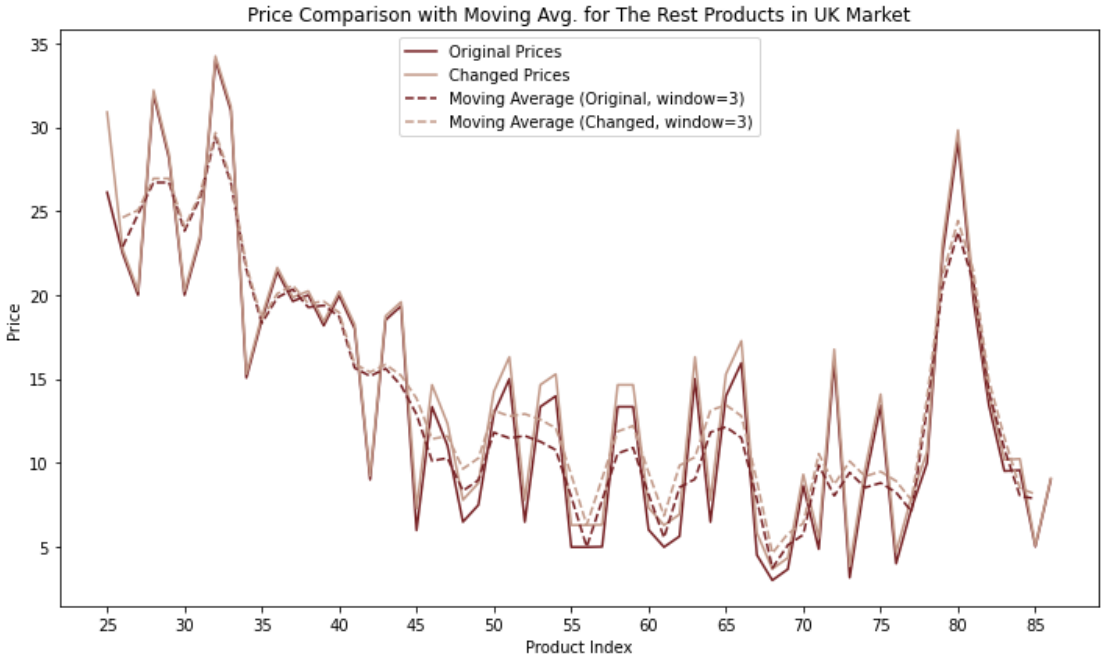


Figure 11: Price Comparison with Moving Avg. for The Merged Companies Products in UK & DE MArket (Local IVS)



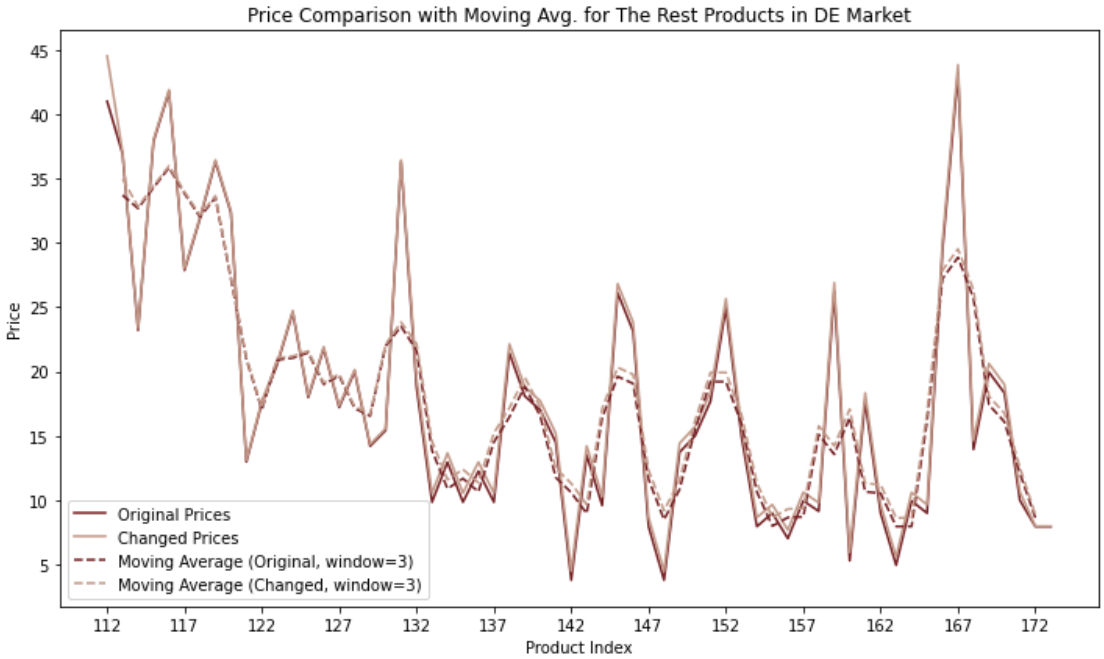
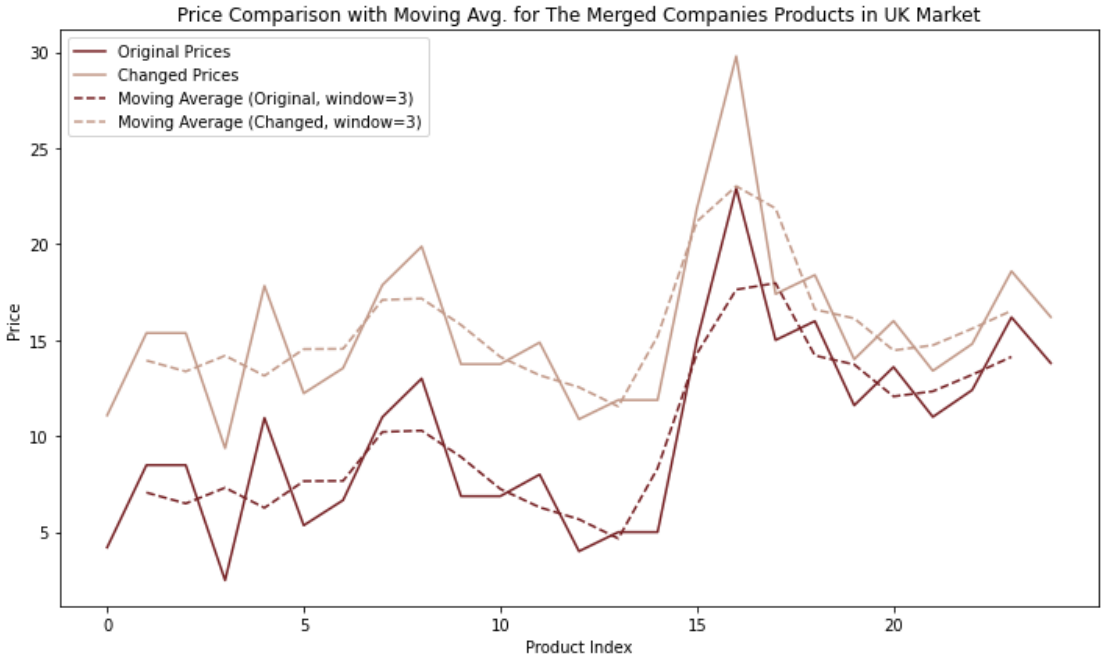


Figure 12: Price Comparison with Moving Avg. for The Rest Products in UK & DE Market (Local IVS)



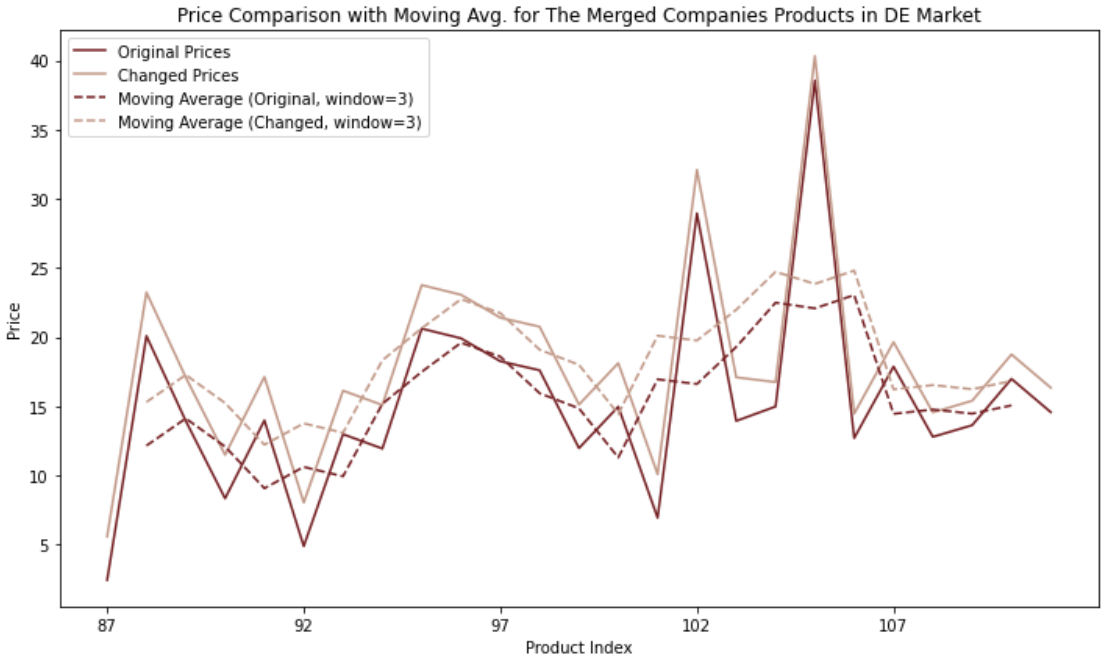
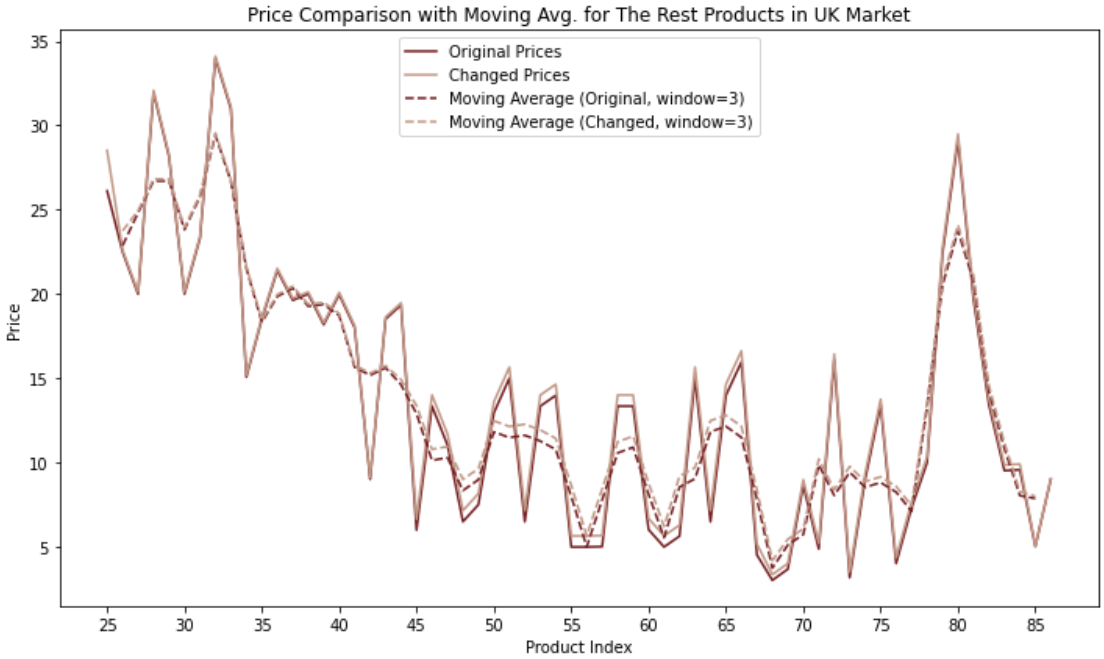


Figure 13: Price Comparison with Moving Avg. for The Merged Companies Products in UK & DE MArket (Quad IVS)



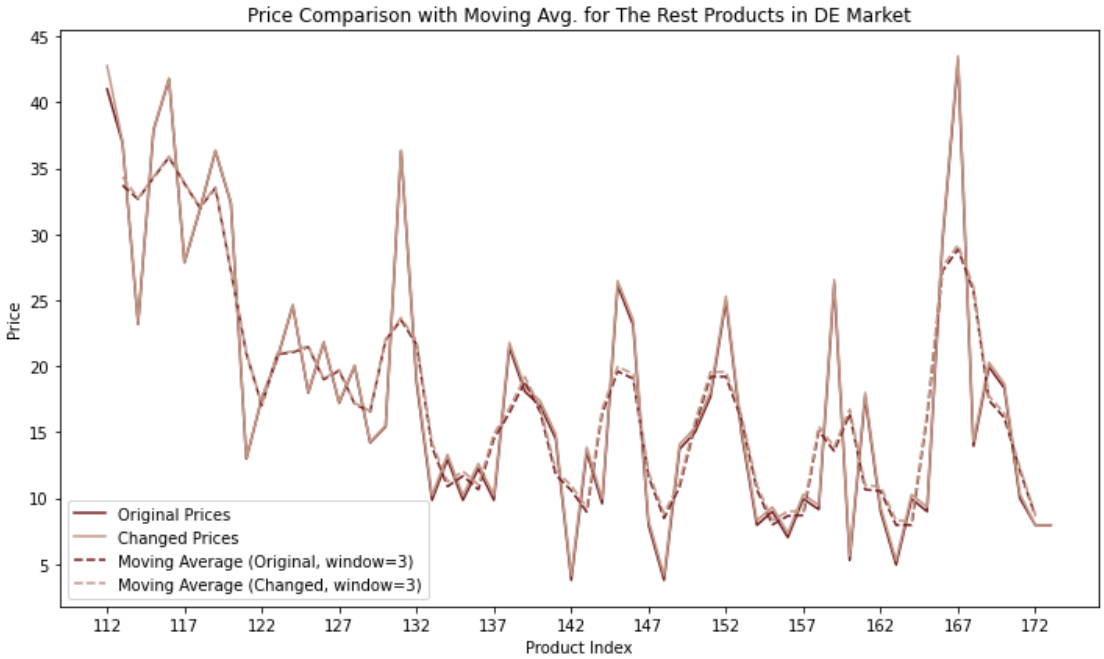


Figure 14: Price Comparison with Moving Avg. for The Rest Products in UK & DE Market (Quad IVS)